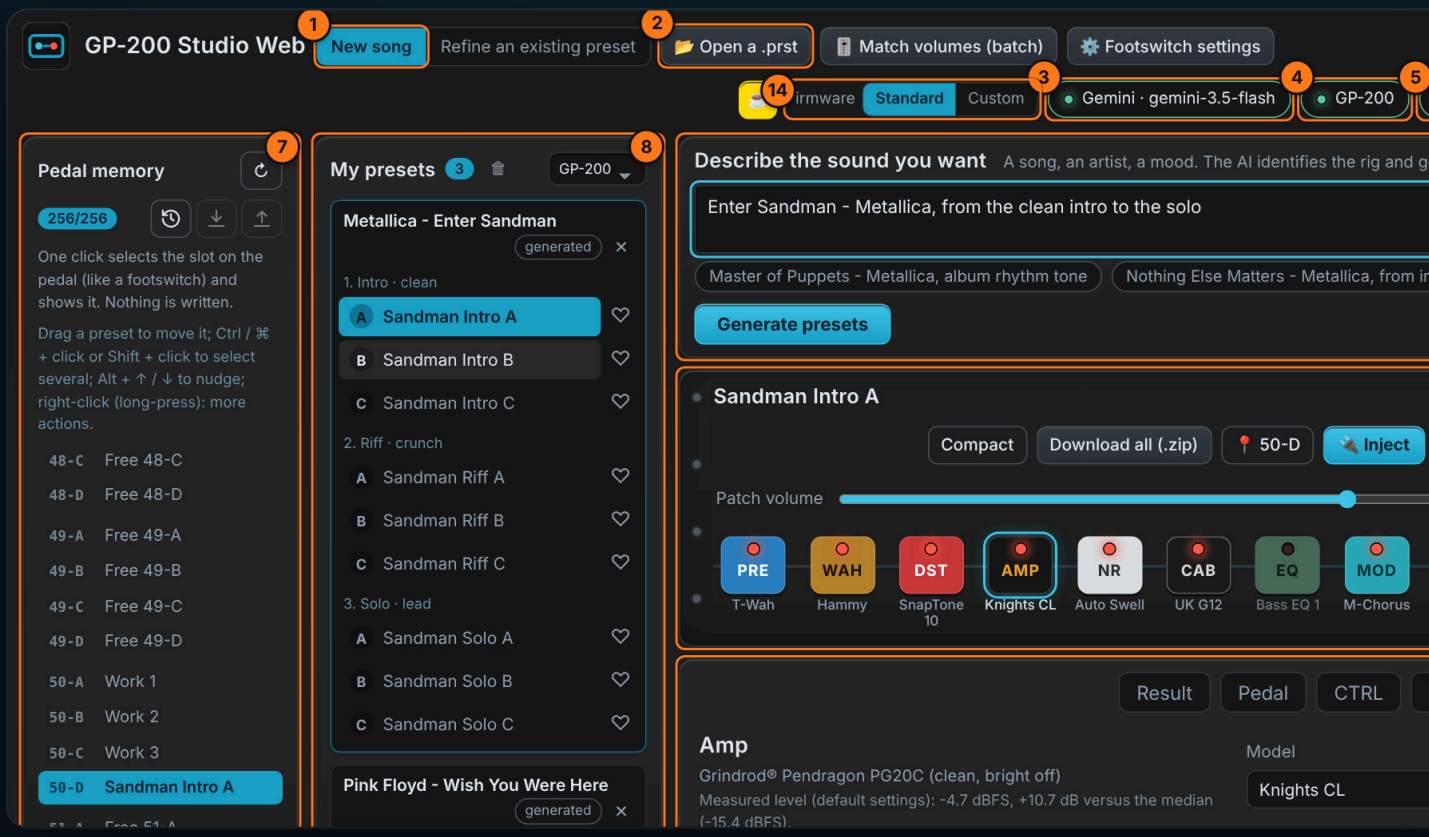
VALETON GP-200 · GP-150 · GP-180

VERSION 0.61.0

GP-200 Studio Web User manual

Generate, open, tweak and organize your presets, send them to the pedal over USB, prepare a gig and match volumes — all in your browser.



Version 0.61.0 · November 9, 2026 · English edition
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This manual describes version 0.61.0 of GP-200 Studio Web. The version number and build date are shown at the bottom of the app: that is how to check you are using the right one.

1 Welcome

GP-200 Studio Web is a preset workshop for the **Valeton GP-200**, **GP-150** and **GP-180** multi-effects pedals. It runs in your browser: nothing to install, nothing to download except the page itself.

What you can do

- **Describe a sound** (“Enter Sandman – Metallica, from the clean intro to the solo”) and get ready-to-play presets: 3 variants for each section of the song.
- **Open your existing presets** (`.prst` files), look at them module by module, tweak them, refine them with AI.
- **Send them to the pedal over USB**, without the official editor, and **tweak the pedal live** while you play.
- **See and organize the pedal memory**: all 256 slots with their names, rearrange them by drag and drop, and import or export presets.
- **Prepare a gig**: keep your favorites ♥, name them, arrange them song by song in the pedal banks, send everything in one go, and add cab filters (Low Cut / High Cut) to all of them at once for loud playing.
- **Bring every preset to the same loudness** (automatic LUFS measurement), so none of them jumps out from one song to the next.

What you need

Browser	**Chrome, Edge or Opera** on a computer (or Chrome on Android, see the chapter “Connecting the pedal”). Safari, Firefox, iPhone and iPad cannot connect to the pedal: you can still create and download presets there.
Address	The page must be opened from its **https** address (for example on GitHub Pages) or from `localhost`.
Pedal	Plugged into the computer over USB. **Only one program at a time** can use the pedal: close the Valeton editor when you use GP-200 Studio.
AI key	Only needed to *generate* and *refine* presets. Everything else (open, send, match volumes, set list…) works without a key.

> **Your privacy.** Everything happens in your browser. Your AI key is sent only to the provider you chose (never to a GP-200 Studio server) and the page loads no external font, script or tracker. The `.prst` files are built on the spot.

How to read this manual

The screenshots carry **numbered orange markers** 1: the number points to the caption just below. Four kinds of boxes come back regularly:

> **Tip**: a shortcut, a good habit

> **Warning**: an action that overwrites something

> **Safety**: what the app does to protect you

> **Not tested on hardware**: verified with a simulated pedal only

A **separate chapter (14)** covers Goaltoday's custom firmware; its pages carry a purple "Custom firmware" band at the top and its boxes are purple. The rest of the manual describes Standard mode.

2 Quick start

Five minutes to go from “nothing” to “a preset sounding in the pedal”.

1. **Open the page** in Chrome or Edge. On first launch, a window offers to connect an AI: paste your key (chapter 4) or click *Continue without AI*.
2. **Plug in the pedal** over USB, switch it on, then click the *Connect the pedal* pill at the top right. Accept the browser’s MIDI permission.
3. **Choose your working slot**: a slot you are willing to sacrifice (for example 50-D). Every preset you audition will be written there. *Never choose a slot that holds a preset you want to keep.*
4. **Describe the sound** in the big field and click *Generate presets*. After a few seconds, the presets appear on the left, and the first one is already in your pedal.
5. **Play, tweak, keep.** Click another preset to hear it, turn a setting (it moves live on the pedal), put a ♥ on the ones you keep, then arrange them with the *Set list* (chapter 10).

> **The working slot is overwritten on every audition.** Nothing is written to the pedal until you confirm that slot. The other slots are never touched, unless you explicitly send a preset or a set list and accept the confirmation.

3 The screen, area by area

On a computer, everything fits on one screen, with no page scrolling. Here is the full screen with a pedal connected.

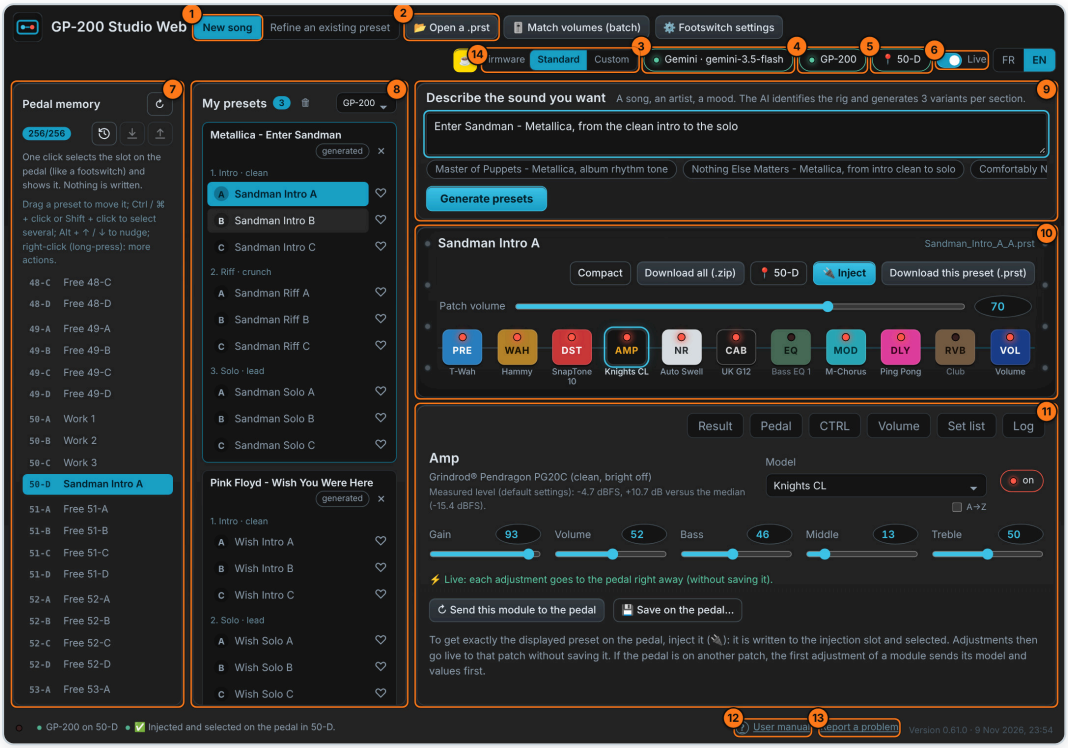


Figure 1 — The workspace (GP-200 connected, memory read).

1. **Mode**: *New song* (generate) or *Refine an existing preset*.
2. **Open a .prst**: loads one or several files (you can also drop them anywhere on the page).
3. **AI pill**: provider and model; a click reopens the connection window.
4. **Pedal pill**: a click connects the pedal (automatic reconnection afterwards); once connected, it opens the *Pedal* tab.
5. **Working slot** (📍 50-D): the slot where every auditioned preset goes.
6. **Live**: the live-editing switch (chapter 6).
7. **Pedal memory**: the 256 slots, with their names (chapter 9).
8. **My presets**: everything you generate or open piles up here; ❤ = favorite; right-click = menu.
9. **Request area**: the field where you describe the sound (or what you want to change).
10. **Rack**: the chain of 11 modules of the displayed preset, with the patch volume above.
11. **Tabs**: *Result*, *Pedal*, *CTRL*, *Volume*, *Set list*, *Log*. They open in a floating window (close: ×, `Esc` or a click outside). Under the tabs: the sheet of the selected module.
12. **User manual** (**?**): opens this manual (PDF) in a new tab, in the language of the interface.
13. **Report a problem** and the version number (bottom).
14. **Firmware: Standard / Custom** (GP-200 only). Leave it on **Standard** unless your pedal runs Goaltoday's custom firmware: Custom mode is described separately, in chapter 14.

The yellow ☕ **Buy me a coffee** button, at the top right, opens the author's Ko-fi page in a new tab. The app is free: it is just an optional link. On a narrower screen (about 1800 px and below) the button is reduced to the ☕ icon (the tooltip gives the details).

> **Tooltips everywhere.** Every button, field, slider and indicator has a tooltip (hover it): what the setting does, its range, its default value. When in doubt, hover. The interface exists in **French and English** (FR / EN buttons at the top right).

4 Connecting an AI

The AI does two things: **generate** presets from a description, and **refine** an existing preset. Without a key, you keep everything else.

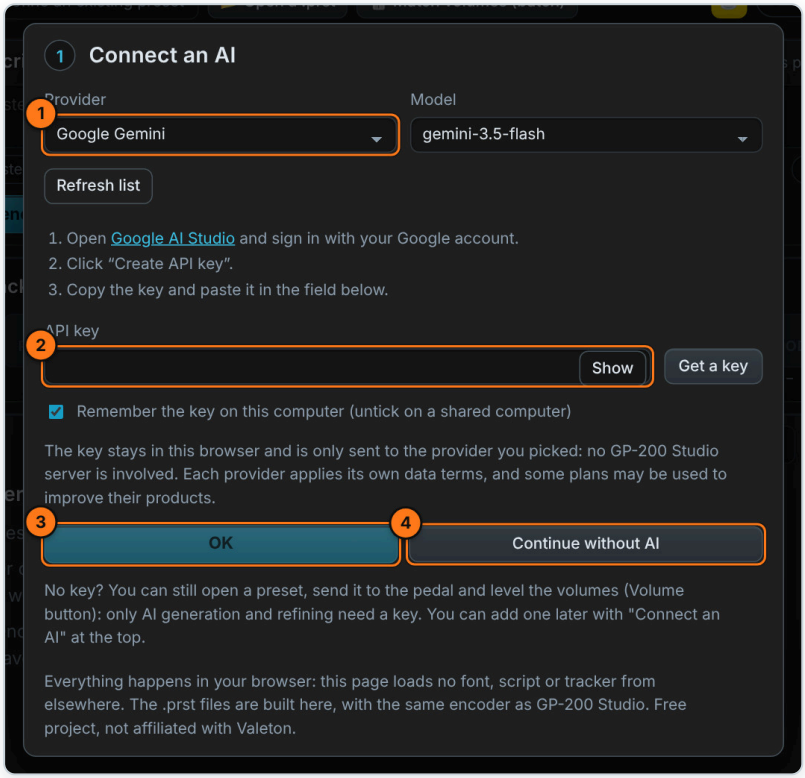


Figure 2 — The connection window (first visit).

1. **Provider and model**: Google Gemini, Anthropic or OpenRouter. (OpenAI and Perplexity do not allow direct calls from a browser.) The *Refresh list* button reads the available models again. One-click OpenRouter connection requires the page to be served over **https**.
2. **API key**: paste it here (the *Show* button reveals it). The *Get a key* links and the 3 steps under the title explain where to create one. The "Remember the key" box keeps it in this browser: untick it on a shared computer.
3. **OK**: confirms. It stays greyed out until there is a key.
4. **Continue without AI**: you can open presets, send them, match volumes, use the set list and the memory. Only "Generate" and "Refine" need a key. You can add one later with the *Connect an AI* pill.

> **Free Gemini key.** Free keys are served after paid keys when the service is busy. If Gemini answers "overloaded" after several tries, the app switches by itself to a *flash-lite* model for the current generation.

> **An "odd-looking" key is not rejected.** The app does not check the format of your key (there are several). Only the provider knows whether it is valid: if it refuses it, you will see "The API key is refused by the provider" at the first generation.

5 Creating presets with AI

Describe a song, an artist, a mood: the AI identifies the rig, splits the song into sections (intro, verse, solo…) and generates **3 variants per section**.

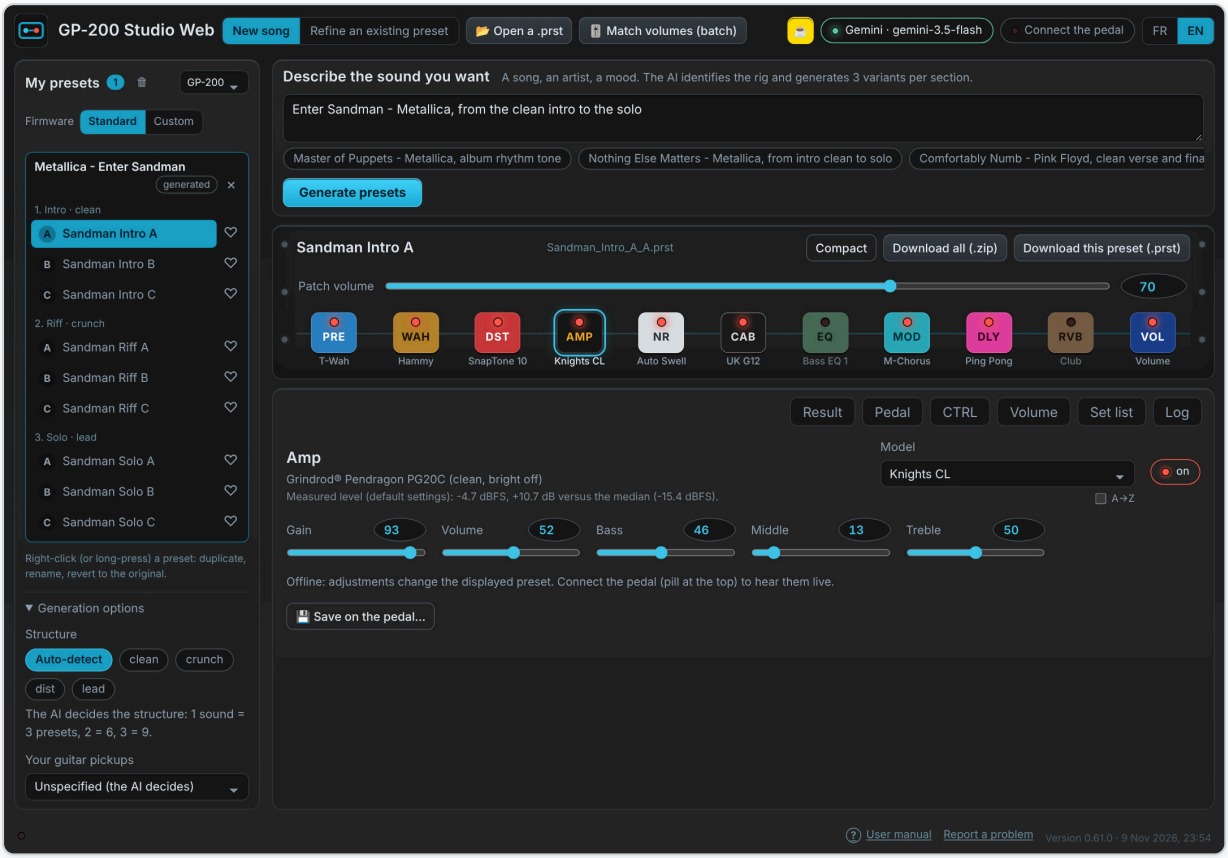


Figure 3 — "Enter Sandman – Metallica": 3 sections × 3 variants = 9 presets, grouped by section in "My presets".

Writing a good request

- Most effective: **title – artist** and, if you like, a precision: "Nothing Else Matters – Metallica, from the clean intro to the solo".
- The **examples** under the field are clickable and fill in the request.
- You can also describe a sound without a song: "warm blues, a bit of reverb, single-coil pickups".

Generation options

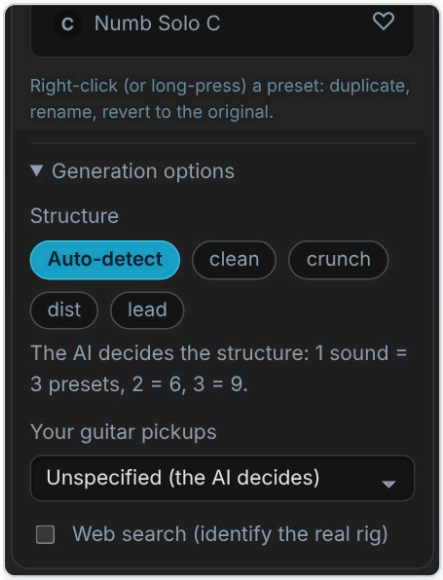


Figure 4 — "My presets" column and options.

Under the preset list, the **Generation options** panel sets what the AI must produce:

- **Structure**: *Auto-detect* (the AI decides: 1 sound = 3 presets, 2 sounds = 6, 3 sounds = 9) or an imposed type: *clean*, *crunch*, *dist*, *lead*.
- **Your guitar pickups**: humbucker, single-coil, P90, active (EMG…), or "Unspecified". The AI takes them into account for gain and EQ.
- **Web search**: asks the AI to look up the real gear used by the artist, when the provider allows it.

During generation, the rack LEDs light up one after the other: this is normal (turned off if your system asks to reduce animations).

The Result tab

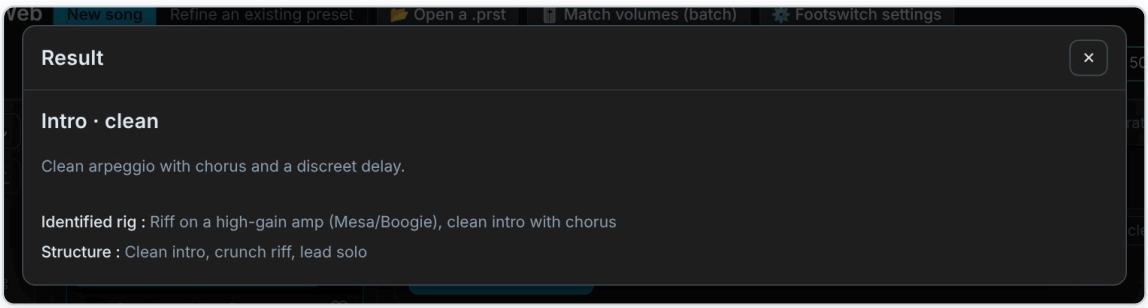


Figure 5 — The *Result* tab: the AI's analysis (section, identified rig, structure).

It sums up what the AI understood: the sound of the displayed section, the identified rig, the structure of the song. After a *refinement*, it also shows the **actual changes** between the old and the new version.

Keeping your presets

- **Download this preset (.prst)**: a file ready to be opened by the Valeton editor or reopened here later.
- **Download all (.zip)**: all the presets of the result, with a detail `.json` per preset.
- Presets stay in "My presets" as long as the page is open. **They do not survive closing the tab**: download or send what matters.

6 Opening and tweaking a preset

The rack shows the full chain of a preset. Each module is adjusted with the mouse, within the pedal's real ranges.

Opening files

Click *Open a .prst* (several files at once is fine) or **drop them anywhere on the page**. An invalid file is reported without blocking the others. Opened presets get the "opened" badge in the list.

The rack and the module sheet



Figure 6 — The rack and the sheet of the *Amp* module.

1. **Patch volume** (0 to 100): the "patch volume" knob of the front panel. It applies to the whole preset, not to one module.
2. **Module**: a first click selects it (focus). One more click on the already selected module turns it on or off. The ← → arrows move from one module to the next.
3. **LED**: module bypass (a single click, always). If the pedal is connected, you see and hear it live.
4. **Model**: the list follows the pedal's real menu for this block (for example 31 models in PRE). The **A→Z** checkbox under the list sorts the groups alphabetically (Acoustic, Bass, Clean, Drive, Hi Gain...; "SnapTone" always stays at the very bottom) and the models within each group; no model ever moves to another group. Unchecked (the default), the list keeps the pedal's order. Only the display changes, and the choice is remembered from one session to the next.
5. **State**: "active" or "bypass".

6. **Settings**: slider or keyboard entry in the pill; at most 6 per row. The *measured level* of the amp or the drive is shown under the title.
7. **Send this module to the pedal**: sends the whole module again (model and all its values).
8. **Save on the pedal…**: opens the *Pedal* tab to write the preset into a slot (chapter 8).

> **Pedal-style blocks, and the Compact button.** Each block of the chain looks like a small pedal: a body coloured by its type (red for distortion, pink for delay, orange for reverb, black for amp…), the LED on top, and the model name written below. The **Compact** button at the top of the rack makes the blocks even shorter; it turns into **Normal** to go back, and the page remembers your choice.

> **Changing the order of the chain (GP-200).** Grab a block of the rack with the mouse and **drag it** to the place you want: a blue bar shows where it will land, and the other blocks shift by one step (example: drag *DLY* right before *MOD*). A simple click keeps its usual role (select the block, then bypass / activate it); **Esc** cancels the gesture, and so does releasing outside the rack. With a finger (phone, tablet), **keep pressing for about 0.3 second** on the block, it lifts, then drag; moving right away just scrolls the page. The displayed preset is changed: the order is written in the file (download, *My presets*, sending to a slot). If the pedal is connected, **Live** is on and pedal listening is active (it tells which patch is current), the pedal takes the new chain as soon as you let go, and the bottom bar says “Chain changed on the pedal (not saved yet)” when it has confirmed it. Like the other live settings, it is temporary until the preset is written to a slot. Dragging is disabled during a generation or a loudness match.

Live editing

When the pedal is connected and the **Live** switch is on, every slider move, model change or bypass goes straight to the pedal. **These settings are temporary**: until you write the preset into a slot, they disappear if you change patch on the pedal. Turn *Live* off so that only the file on screen changes.

> **The other way too.** If you turn a knob on the pedal’s front panel, or press a module footswitch, the page follows: the slider, the LED and the file update. The patch volume is followed the same way. **Changing the model** of a block (amp, distortion, modulation, delay, reverb, cab) on the pedal is followed too: the page then re-reads the patch and shows what the pedal really holds. This re-read never writes anything to the pedal, and an opened or generated preset is not replaced.

> **The ↻ symbol** next to a setting means it is “silent live”: the pedal only takes it into account when the preset is loaded (a few Sync, Trail, Spread and mode settings). Send the module again or write the preset to hear it.

> **Goaltoday's custom firmware.** If your pedal runs the custom firmware (universal blocks, NAM, scenes, Dual Chain), everything about it is described separately in **chapter 14**. This chapter only describes **Standard** mode, the one the app always starts in.

The right-click menu on a preset

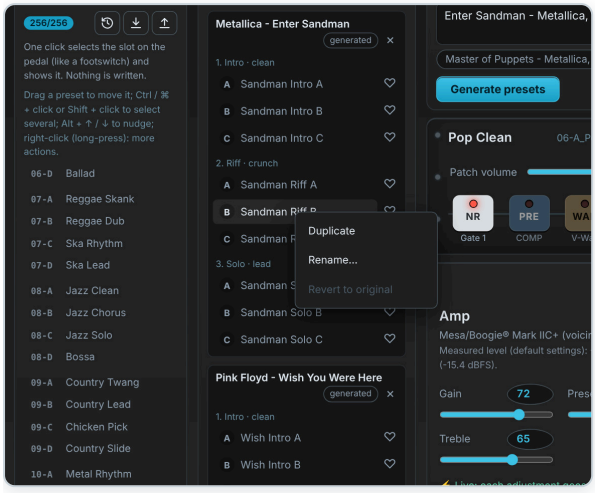


Figure 7 — Right-click on a preset.

In “My presets”, a **right-click** (long press on touch, or the `Menu` / `Shift`+`F10` key) offers:

- **Duplicate**: creates an independent copy right below (“Name 2”, “Name 3”…), to try a boost without touching the original.
- **Rename…**: changes the name (16 characters at most on GP-200, 13 on GP-150/180, accents removed) and the file name.
- **Revert to original**: restores the preset as it was when created (for a copy: as it was at the time of duplication, matched volume included).
- **Delete this copy**: on copies only, never on a generated or opened preset.

Copies can be favorited, exported, volume-matched and sent like the others. The menu is unavailable during volume matching or a send.

7 Refining a preset

A preset is almost right? Tell the AI what is wrong, in plain English. It corrects the selected preset and adds the result to the list, **the original is kept**.

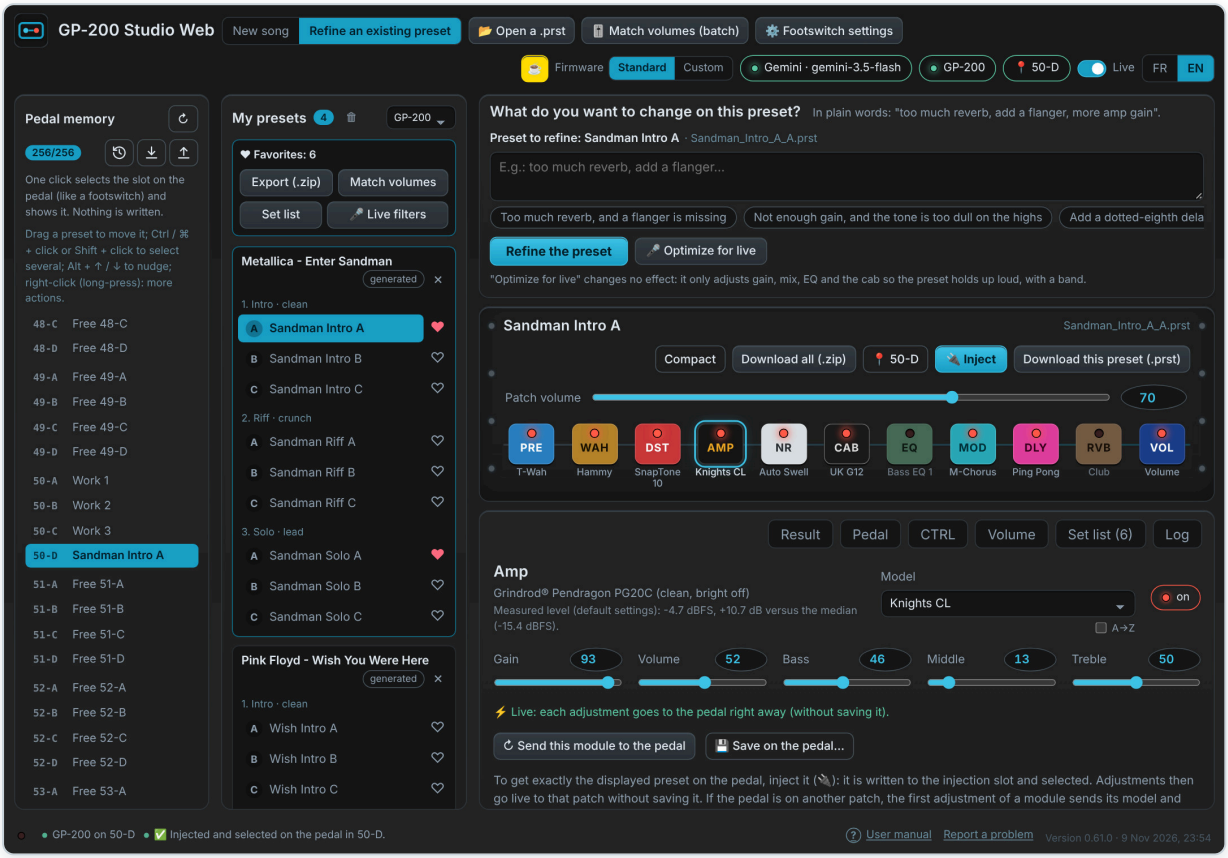


Figure 8 — The *Refine an existing preset* mode.

- Choose the **Refine an existing preset** mode, select the preset in the list (generated, opened or already refined), then write: “too much reverb, it needs a flanger”, “not enough gain, the sound is dull in the highs”, “add a dotted-eighth delay”.
- **Refine the preset** sends the request. The new preset arrives in the list (“refined” badge) and the *Result* tab shows the **actual changes**.
- 🎤 **Optimize for live** changes no effect: it only adjusts gain, mix, EQ and cab to play loud, in a band. To add the cab filters and set the noise gate of the high-gain presets of all your favorites in one go, without AI, see “Live filters” in chapter 10.

> **Your settings are not lost.** Refining keeps the CTRL table and the patch volume of the original file, as well as the models already in the preset (even outside the current menu). Only a *new* model that the pedal does not offer in this block is refused and sent back to the AI for correction.

8 Connecting the pedal

Connection, working slot, listening to the pedal and sending presets.

Connecting

1. Close the Valeton editor if it is open (only one program at a time can use the pedal).
2. Plug the pedal in over USB, switch it on.
3. Click the *Connect the pedal* pill. The browser asks for permission to access MIDI devices **with SysEx**: accept. On later visits, reconnection is automatic.
4. The pill turns green and shows the recognized model (GP-200, GP-150 or GP-180). On connection, the page **reads the current patch** and displays it.

Choosing the working slot

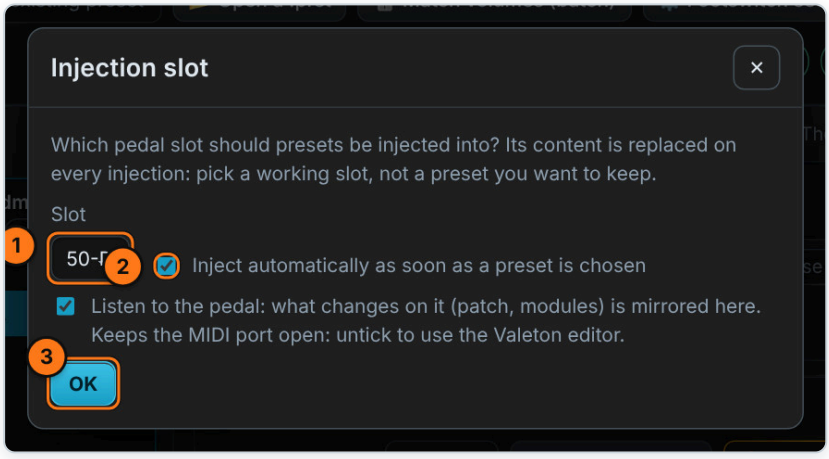


Figure 9 — The “Injection slot” window, opened as soon as you connect.

1. **Slot list** (from 01-A to 64-D, with the name of the preset it holds, for example “50-D Smell Disto”): the working slot. The name shown is that of the preset that will be overwritten; it appears once the “Pedal memory” has been read. The 📍 pill in the header (here 50-D) reminds you of it and reopens it at any time.
2. **Inject automatically** as soon as a preset is chosen (click in the list, generation, opening, refinement result). Without this box, the 🔌 *Inject* button sends the preset on demand.
3. **OK**: *nothing is written to the pedal until you confirm*.

The **Listen to the pedal** box (ticked by default) keeps the MIDI port open so the page follows what happens on the pedal: patch change, module toggle, knob turned or model changed on the front panel. **Untick it to use the Valeton editor**, which does not share the port.

> **The working slot is rewritten on every selection.** Choose a slot you do not need. Injection writes the preset into this slot and then selects it: the pedal loads it from its memory, so every module has the right model and live settings target the right parameters.

The Pedal tab: sending presets



Figure 10 — The *Pedal* tab.

1. **Read the current patch**: reads on demand the patch currently loaded on the pedal (unsaved settings included) and adds it to the list with the "pedal" badge. It is **read-only**: this patch is never sent back automatically.
2. **Slot list** for the destination, with the name of the preset currently in each ("12-B Zombie Clean": that is the one that will be overwritten).
3. **Send this preset to 12-B**: writes the displayed preset into the chosen slot, after confirmation ("What you are about to overwrite").
4. **Send the 3 variants**: writes the displayed section (A, B, C) into three consecutive slots.

At the bottom, the *What the pedal says* panel keeps the last messages received in hexadecimal: useful for a problem report.

> **If the pedal changes patch.** Live settings are suspended (nothing is sent to another patch) and a banner offers *Send the preset again*. If the pedal reloads the working slot while you had settings that were not injected, a banner points it out.

> **Discreet messages.** When all is well ("GP-200 on 50-D · patch volume 57", "Injected and selected..."), the information stays in the bottom bar. Alerts that need an action stay near the rack.

On Android (USB OTG cable)

Chrome for Android + an OTG adapter and the pedal's USB-B cable (or a USB-C → USB-B cable) give the same sends and the same live editing. The page must be opened **from its https address**: opened from an attachment or a downloaded file, Chrome refuses MIDI. Sending is paced more cautiously there.

9 The pedal memory

A column on the left lists the **256 slots** (01-A to 64-D) with the name of the patch stored in each. You can browse them... and rearrange them.

Reading and browsing

- Reading starts a few seconds after connection and takes about 6 seconds. The counter (**256/256**) shows progress. If this first reading fails right after connecting, the page tries **once** again by itself, about 4 seconds later. The ↻ button reads everything again. The names read last time appear **right away** at launch (greyed, “to re-read”) and are then checked in the background. If the pedal answers slowly (this can happen after saves to its memory; restarting it usually fixes it), the page says so under the counter: it then reads only the names around the current patch and those you scroll to; ↻ reads all 256 (allow about a minute).
- **A click on a slot** selects it on the pedal (like with your foot), then the page reads it and shows it in the rack.
- After an injection or a set list send, the affected slots are read again right away. On reconnection, names already read stay displayed (greyed) and are then read again.
- Reading is **read-only**: no write is ever sent by simply browsing.

Rearranging by drag and drop (GP-200)

A single preset. Want to put 40-A in place of 42-B? Grab it and drag it onto it: the target row is underlined, it takes that place and the others shift by one slot (handy when the order of the songs in your set list changes). The move is only a **preview**: the list shows the result and nothing is written. You can chain as many moves as you like; **only when you click “Apply to the pedal...”** does a confirmation window show the summary of everything, before it is all written in one pass. A simple click, without dragging, still loads the preset on the pedal.

Several presets. Same principle for a block: Ctrl / Shift + click to select, then drag the group. The move is a **plan**, only displayed: you see the result before the pedal is touched (figure 11 shows one: two presets, 40-A and 40-B, dropped on 42-B).

Mistakes are free. “Undo last move” undoes the last move; each click undoes one more. “Cancel all” goes back to the pedal’s state in one click.

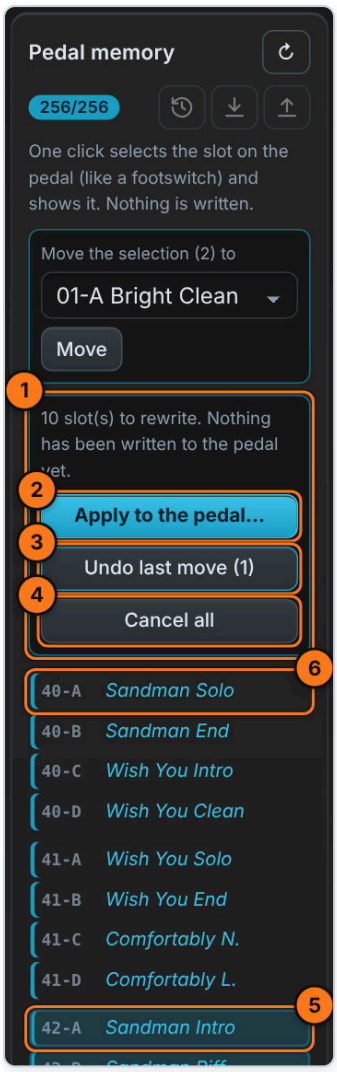


Figure 11 — A pending plan.

1. **Plan banner**: "10 slot(s) to rewrite. Nothing has been written to the pedal yet."
2. **Apply to the pedal…**: starts writing (after confirmation).
3. **Undo last move** (the number in brackets is the number of moves in the plan): undoes the last one, then the one before…
4. **Cancel all**: abandons the plan, everything goes back as on the pedal.
5. **Destination** of the moved block (here 42-A and 42-B receive "Sandman Intro" and "Sandman Riff").
6. **The others shift**: italic lines with a left border are those that will be rewritten; hovering shows where each content comes from.

You can chain several drag-and-drops: each one is added to the plan, and a single send applies the whole.

Moving several presets at once

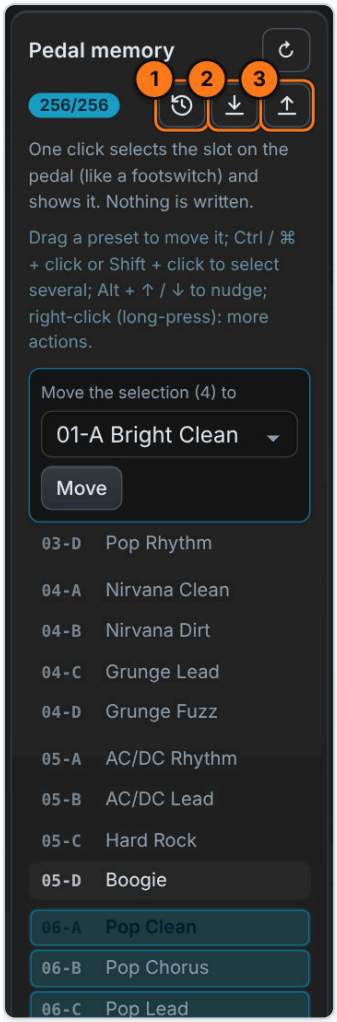


Figure 12 — Multiple selection. At the top, the three icons: 1 Restore, 2 Import, 3 Export.

- `Ctrl` (or `⌘`) + click: adds or removes a slot.
- `Shift` + click: selects a range.
- `Shift` + click with no previous click: the range starts from the slot loaded on the pedal.
- Drag one of the selected slots: **the whole block** moves, keeping its order (the image following the cursor shows the number of slots). Example: select 50-A → 54-D and drag them onto 01-A: they become 01-A → 05-D and the rest shifts.
- To reach a distant slot, bring the cursor near the top or bottom of the list: it scrolls by itself, faster and faster near the edge.
- **Move the selection to** [slot list, with names] **Move**: under the list, places the selected block where you want without dragging (same plan, same safeguards).
- `Alt` + `↑` / `↓`: shifts the selection by one step, from the keyboard (also the way to do it on a touch screen, where dragging is not supported).

During a plan, a simple click selects without loading anything (the displayed content is no longer the pedal's).

Applying the plan

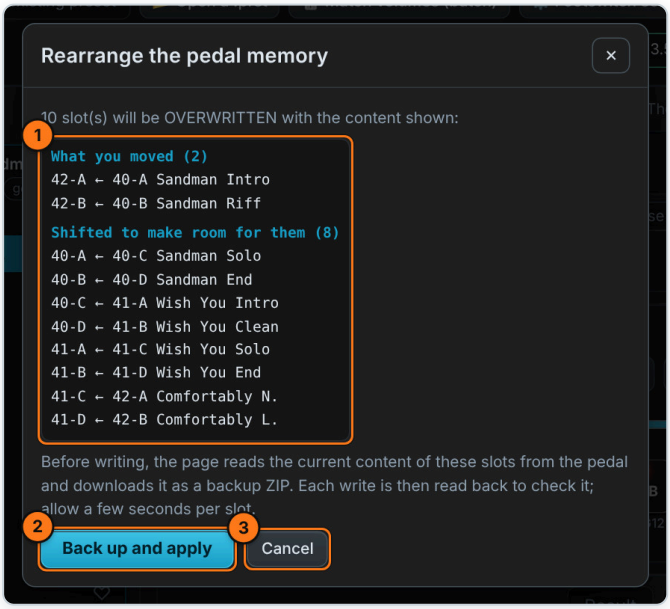


Figure 13 — The confirmation before writing.

1. **The exact list of overwrites**: “42-B ← 40-A P156”... each slot with the content it will receive. It has two groups: first **what you moved**, then the presets **shifted to make room for them** (they take the freed slots).
2. **Back up and apply**.
3. **Cancel**: nothing has been written.

Clicking *Back up and apply* runs five steps, with a progress bar:

1. **Reading** of all the slots involved, with a name check. If one of them changed since it was displayed, **everything stops before the slightest write** (read again with ↻, then start over).
2. **ZIP backup** (`pedal_backup_YYYYMMDD_HHMM.zip` , one `.prst` per slot) downloaded *before* the first write.
3. **Writing**, slot by slot. Allow a few seconds per slot; do not unplug the pedal.
4. **Verification read** of every written slot; those that differ are flagged “To check”.
5. The pedal moves to the first modified slot.

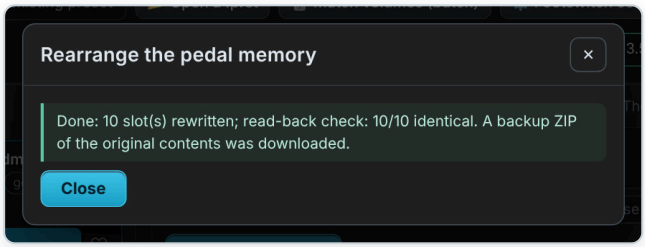


Figure 14 — End of the operation: 10 rewritten, 10 read back identical, backup downloaded.

> **If something goes wrong.** If the cable is unplugged during the send, the plan is abandoned and the message says how many slots were written and which ones (the one in progress is marked “(?)”). The memory is read again, and the ZIP backup lets you restore everything: see below.

The three icons of the column

To the right of the 256/256 counter, at the top of the column (marks 1 to 3 of figure 12), three **icons** give access to file exchanges with the pedal memory. Hover over them: a tooltip explains each one, and why it is greyed out when it is. They are enabled once the 256 names are read and no send or move plan is in progress. GP-200 only.

Icon	Name	What it does
circular arrow with a clock	**Restore...**	Writes a backup (ZIP, or `.prst` files named `07-B_name.prst`) back to the original slots. See “Restoring a backup” below.
arrow pointing down to a tray	**Import .prst files...**	Writes one or more `.prst` files into the memory, starting at the selected slot, then loads the first one on the pedal.
arrow pointing up out of a tray	**Export the selection**	Downloads the selected presets: one `.prst` file if a single one, a ZIP if several. Read only.

Import — the icon is greyed out until a slot is selected:

1. Click the slot where the **first** preset should land (with Ctrl / Shift + click, the topmost slot of the selection counts).
2. Click the **Import** icon and choose one or more valid GP-200 `.prst` files. Several files go into **consecutive** slots, in alphabetical order of their names (name them `01_…` , `02_…` to choose the order).

3. If a file is not a valid GP-200 `.prst` (size, checksum), or if the batch goes past the last slot (64-D), **everything is refused**: nothing is written and the faulty file is named.
4. A window lists each slot, the preset it will receive and what it *currently* holds. Nothing is written before **Back up the current state and write**.
5. The page downloads the current state as a `before_change_…` ZIP, writes, reads every slot back to check it, then **loads the first imported preset on the pedal**.

Export — select one or more presets (Ctrl / Shift + click), then click the icon. Files are named like the backups (`26-A_Name.prst`): they can be imported or restored again. Nothing is changed on the pedal. For the whole memory at once, use the backup ZIP made before a rearrangement.

> **Copying presets from one pedal to another.** Export the selection, open the page with the other GP-200, select the first destination slot, then import the files extracted from the ZIP.

Restoring a backup

The ZIP downloaded before each rearrangement holds one `.prst` per slot, named after its slot (`07-B_PresetName.prst`). To go back:

1. In the “Pedal memory” column, click the **Restore…** icon (circular arrow with a clock, next to the 256/256 counter; enabled once the 256 names are read and no plan is pending).
2. Choose the **backup ZIP** (or directly `.prst` files whose name starts with the slot). Files with no slot in their name, a wrong size or a bad checksum are ignored and reported.
3. A window lists each slot, the preset it will receive and what it *currently* holds. Nothing is written until you click **Back up the current state and restore**.
4. The page reads the current state, downloads it as a `before_restore_…` ZIP (so a restore can itself be undone), writes, then reads every slot back to check it.

> **Recent feature.** The move was verified against a simulated pedal that really keeps the content of its 256 slots (byte by byte). Not yet on a real pedal: keep the ZIP backup at least the first time. Available on GP-200 only.

The slot menu (right-click)

A **right-click** on a slot in the column (or a **long press** on a touch screen) opens an action menu, like on “My presets”. If several slots are selected (Ctrl / Shift + click) and you right-click one of them, the menu applies to the whole selection when it can.

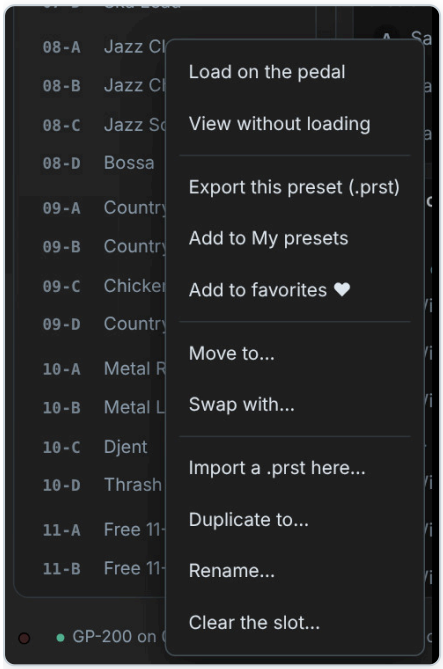


Figure 15 — The slot menu.

Read-only — nothing is written to the pedal:

- **Load on the pedal**: like a click (the pedal selects the preset, the page shows it).
- **View without loading**: reads the stored preset and shows it in the rack, *without changing the pedal’s current preset*.
- **Export this preset (.prst)**: downloads the file, named `26-A_Name.prst` (readable by “Restore...”). With several slots selected: **Export the selection as ZIP**.
- **Add to My presets**: copies the preset into your list, where you can duplicate, rename or edit it.
- **Add to favorites ♥**: same as the previous line, but the preset(s) are also marked ♥ (the whole selection at once). Handy to build your favorites list, then click “Match volumes” in the favorites bar.

Plan — a preview in the list, nothing is written until “Apply to the pedal...” (and “Undo last move” works):

- **Move to...**: pick the slot from the list (number, letter and name of the preset it holds); for a selection, the whole block moves.
- **Swap with...**: two slots swap their contents.

Write — each one asks for confirmation, first downloads the current state as a `before_change_…` ZIP, then reads the slot back to check it:

- **Import a .prst here...**: a valid GP-200 file replaces the slot’s content (for several files at once, use the **Import** icon).
- **Duplicate to...**: copies the slot to another one (which is overwritten).
- **Rename...**: changes the name only; the rest of the preset is rewritten identically.
- **Clear the slot...**: writes a neutral preset named “Empty” (all effects off).

> **While moves are pending**, only “Move to...” and “Swap with...” stay active: the screen no longer shows what the pedal really holds, so apply or cancel first. A greyed-out item tells you why on hover. For a rename, the preset is read again just before writing: only the name changes, even if the slot was modified in the meantime.

10 Favorites, set list and gigs

You are preparing a gig: you audition dozens of presets, keep a few, name them, and arrange them in the pedal in set list order.

Keeping with the heart ❤

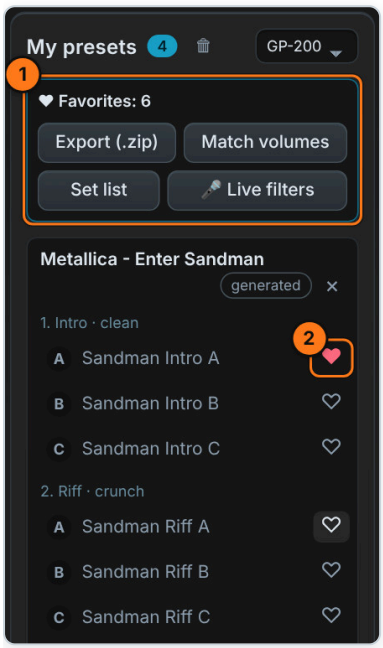


Figure 16 — The favorites bar.

1. **Favorites bar**, which appears as soon as a preset is kept: **Export (.zip)** (one `.prst` per favorite, numbered 01_, 02_... in set list order), **Match volumes** (opens the Volume window with the favorites only), **Set list** and 🎤 **Live filters** (see below).
2. ❤ to the right of each preset: a click keeps or removes it. Keeping a preset does not change the selection: you keep auditioning.

> **The favorites list is kept only in the tab.** If you close the page, it disappears: export (ZIP) first.

Keeping presets read from the pedal, and clearing the list

- A click on a slot of the “Pedal memory” column shows that preset in “My presets” (“pedal” label). The next click **replaces** it... unless you marked it ❤: it then stays in the list and the next preset is added below it. You can browse the memory and tick ❤ as you go; only unmarked presets are replaced.
- To mark several pedal presets at once: select them (Ctrl / Shift + click), right-click, **Add to favorites ❤**. Nothing is written to the pedal. If the preset already shown (“pedal” label, unmodified) is part of the selection, that entry itself gets the ❤: no copy. The same preset added a second time does give a copy (handy to match one in Lead mode and the other in Normal).
- The 🗑 button at the top of “My presets” clears the whole list, favorites included, without touching your files or the pedal. It asks for confirmation: a first click turns it into “Clear N?”, a second click confirms (without a second click it reverts by itself after 4 s). It cannot be undone.

The Set list window

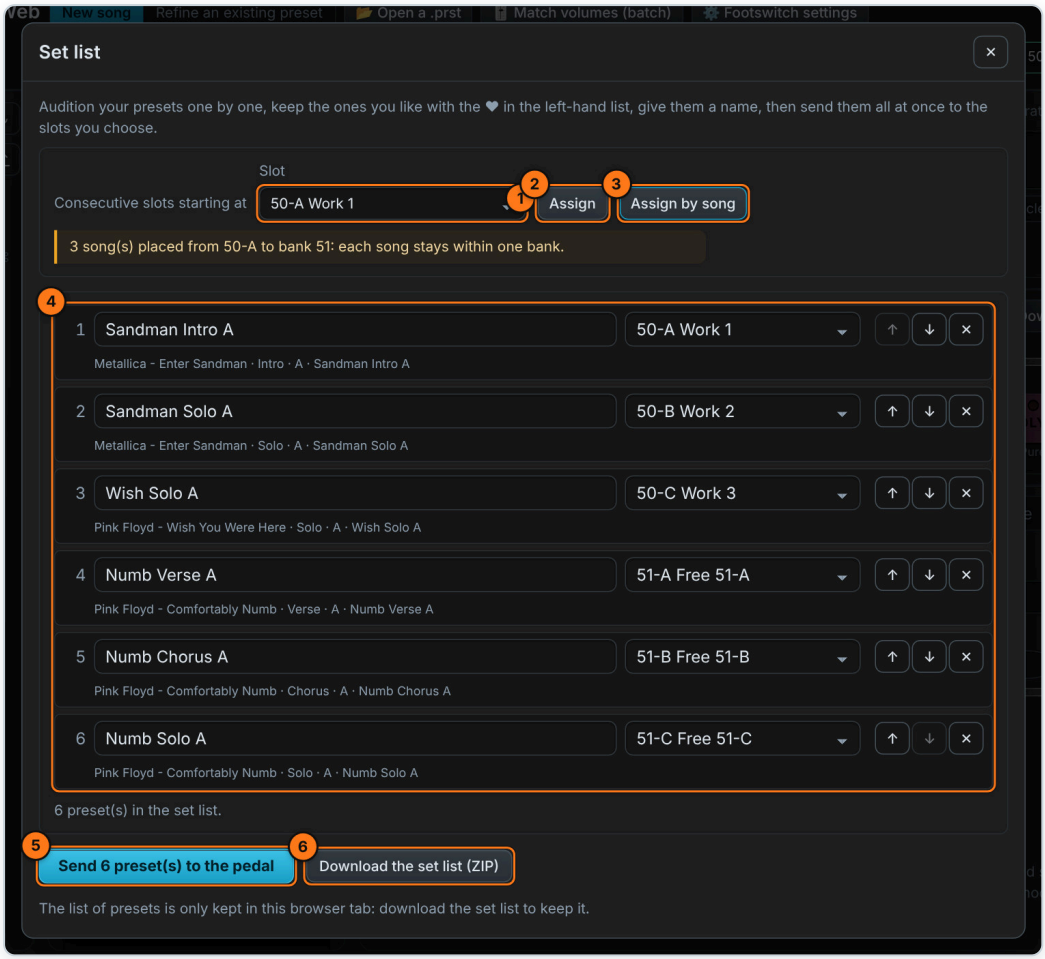


Figure 17 — The set list, arranged “by song” starting at 50-A.

1. **Starting slot** (list with the names of the current presets): the first slot to fill.
2. **Assign**: *consecutive* slots starting from that one (50-A, 50-B, 50-C, 50-D, 51-A...).
3. **Assign by song**: each song stays in *a single bank* (see below).
4. **The list**: one name per preset (16 characters, accents replaced by the base letter: “Café” becomes “Cafe”), a slot (list with the name of the preset currently in that slot; “–” = no slot), and the ↑ ↓ ✕ buttons to sort or remove.
5. **Send N preset(s) to the pedal**: asks for a confirmation, then writes everything.
6. **Download the set list (ZIP)**: to keep it (the list is not kept after the page is closed).

“Assign by song”: one song, one bank

On stage, switching between presets of the same song without changing bank is much safer. The button groups the favorites by song (in order of first appearance) and fills letters A to D of each bank. **A song never straddles two banks**: if it does not fit in the remaining letters, it starts in the next bank.

Case	Result starting at 50-A
1 preset, 1, 1, 1 (four single-preset songs)	50-A, 50-B, 50-C, 50-D: the four letters of a bank are used
1, 1, 1, 2, 3	50-A, 50-B, 50-C / 51-A, 51-B / 52-A, 52-B, 52-C
2, 1, 3 (as in Figure 17)	50-A, 50-B / 50-C / 51-A, 51-B, 51-C
A song with more than 4 presets	it takes whole banks (4 per bank)

Sending



Figure 18 — The overwrite confirmation.

1. **The list of overwritten slots**, with what they will receive (“50-A ← Sandman Intro A”).
2. **Overwrite and send**: written one by one, with a progress bar; the pedal moves to the first preset at the end.
3. **Cancel**.

> **Checks before sending.** *Blocking*: empty name, missing or invalid slot, two presets on the same slot, more than 256 presets. *Warnings*: modified name (accents removed), two identical names, slot equal to the working slot (it will be rewritten on every selection). In case of interruption (cable unplugged…), the message says how many presets were written and which ones.

> **Your files are never modified.** The name you type exists only in the copy sent to the pedal (16 name bytes + recomputed checksum, nothing else).

Live filters: getting all favorites ready for loud playing

On stage, a cab with no filtering lets through a boomy low end and a fizzy top end. The 🎤 **Live filters** button in the favorites bar sets two filters on the cab (CAB) of every favorite: **Low Cut 80 Hz** and **High Cut 8 kHz**. If you wish, it also sets the **noise gate** of heavily distorted presets: at high volume, a gate that is too low lets the hiss through and invites feedback. It uses no AI: the result is always the same, and it is instant.



Figure 19 — The “Live filters: favorites” window.

1. **Noise gate**: the box (ticked by default) and the target threshold (60 by default, from 1 to 100). Untick the box to set the cab filters only.
2. **The list of your favorites**, with what will be done to each one: “changed” (values before → after), “already fine” or “skipped”. The gate line also tells you its position: “at the head” or “at the end”.
3. **Each in its original slot**: only the presets read from the pedal that need changing are rewritten, in the slot they came from. The other favorites are left aside.
4. **One after the other, starting at…**: you pick a starting slot (list “10-A Preset name”). All favorites are laid out one after the other, in set list order, **changed or copied as they are**: you get your complete list back, with no gaps. With 35 favorites and a start at 10-A, they fill 10-A to 18-C.
5. **Continue**: opens the summary. Nothing is written until you confirm.

Cab setting	What the button does
No Low Cut	set to 80 Hz
Low Cut already set (even above 80 Hz)	never touched
No High Cut, or above 8 kHz	brought down to 8 kHz
High Cut already lower (6 kHz, 5 kHz…)	never raised
Cab switched off, or no cab	filters skipped: they would have no effect (the noise gate can still be set)

The **noise gate** is only set on **high-gain** presets: a Hi Gain amp with Gain at 50 or more, any other amp (except bass and acoustic) with Gain at 65 or more, or a distortion / fuzz that is on with Gain at 60 or more.

Noise gate of a high-gain preset	What the button does
Gate on, threshold lower than 60	threshold raised to 60
Gate on, threshold already 60 or more	never lowered
Gate off	switched on, threshold raised to 60 if it was lower
Another effect in the gate slot	not touched, flagged in the list
Capture-type amp (SnapTone)	not evaluated, not touched
Clean or light crunch sound	gate never touched

Only the threshold changes: the gate’s other settings stay identical. The gate can sit **at the head** of the chain (it listens to the guitar before the amp) or **at the end** (it listens after the amp): the same threshold does not act the same way there. The app does not compensate and never moves the gate; it only tells you its position. A high threshold also cuts the tail of notes: it suits heavily distorted sounds. It reduces the hiss between notes; against real feedback while you play, you also need to move away from the amp or lower the gain a little.

Nothing else changes in the preset: name, models, effects, preset volume and pedal assignments stay identical. A preset that is already fine is not rewritten (in “each in its original slot” mode) or simply copied as it is (in “one after the other” mode).

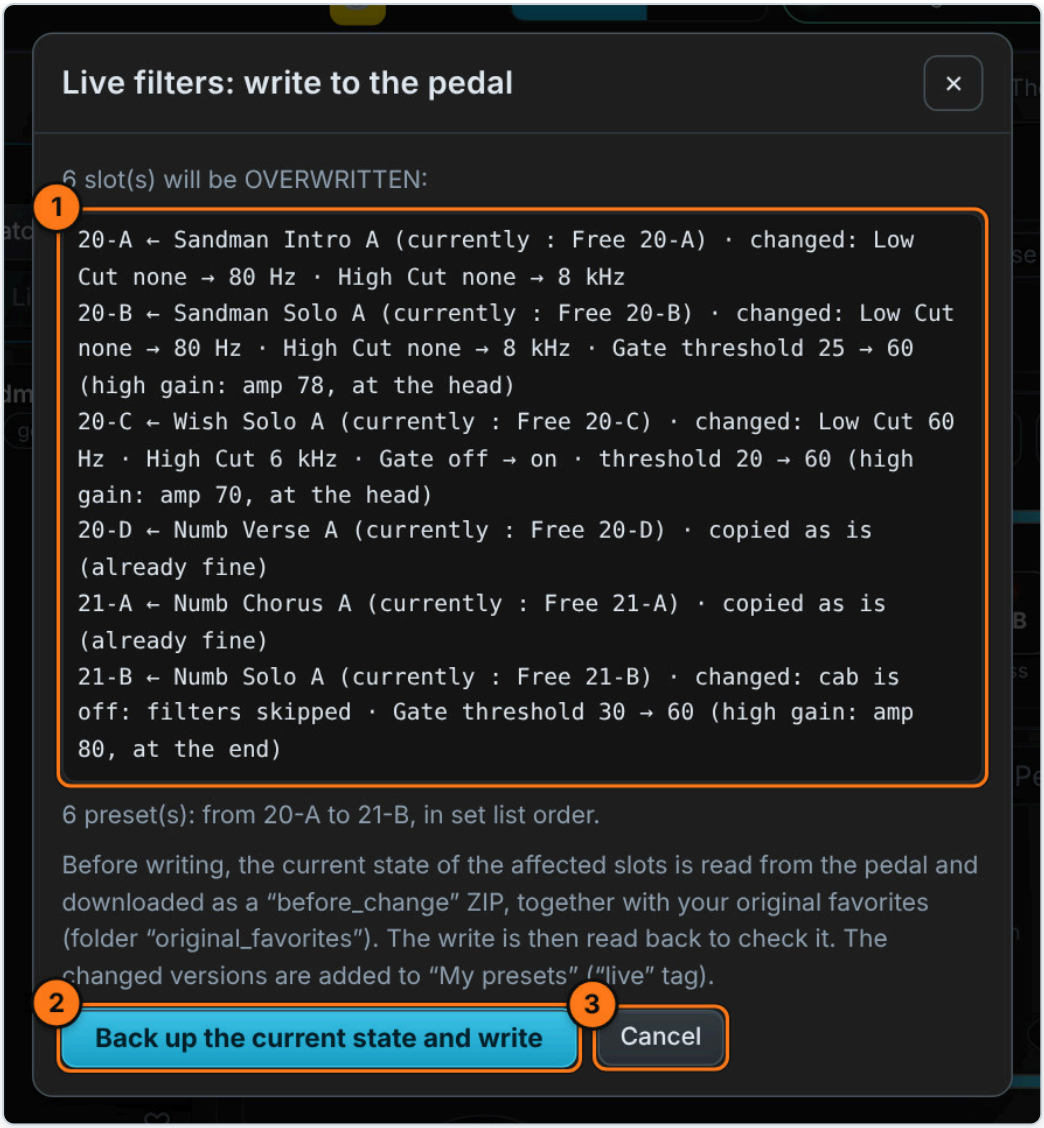


Figure 20 — The summary before writing.

1. **The slots about to be overwritten**: "slot ← preset", with the name currently in the slot and what changes.
2. **Back up the current state and write**.
3. **Cancel**: nothing has been written.

- Before writing, the app reads the current content of the affected slots from the pedal and downloads a `before_change` ZIP: the previous state of each slot, plus your original favorites in the `original_favorites` folder.
- After writing, every slot is read back to check it. If a mismatch is reported, no version is added to the list.
- The changed versions join "My presets" with the "live" tag (files `name_live.prst`). Your original favorites are not touched.
- To put everything back as it was, use **Restore a backup** ("Pedal memory" column) with the ZIP.

> **Safeguards.** The **Continue** button stays greyed out, with a message saying why, if the list runs past the end of the memory (the last slot is 64-D), if the range contains the working slot, if a favorite is unreadable, or if there is nothing to change. In "each in its original slot" mode, a favorite that came from the working slot, or that has the same original slot as another favorite, is left aside. If the write is interrupted (cable unplugged…), the message says how many presets were written, and the ZIP lets you restore everything.

11 Matching volumes

A preset that is too loud or too quiet ruins a run of songs. Here, you play a chord on a loop: the app listens to the pedal's output, measures the loudness in **LUFS** (the unit used by YouTube and Spotify) and adjusts by itself until it reaches the target (GP-200: −14 LUFS for Normal, −11 for Lead, i.e. 3 dB apart; GP-150/GP-180: −16 and −11; adjustable preset by preset).

What you need

- The pedal connected over USB **and its USB audio input** (it shows up as a "GP-200" sound card): the browser asks for permission to use the microphone / audio input.
- Your guitar plugged into the pedal.
- The tab **visible during the measurement**: the browser slows down hidden tabs.
- Chrome or Edge on a computer.

Preparing the batch

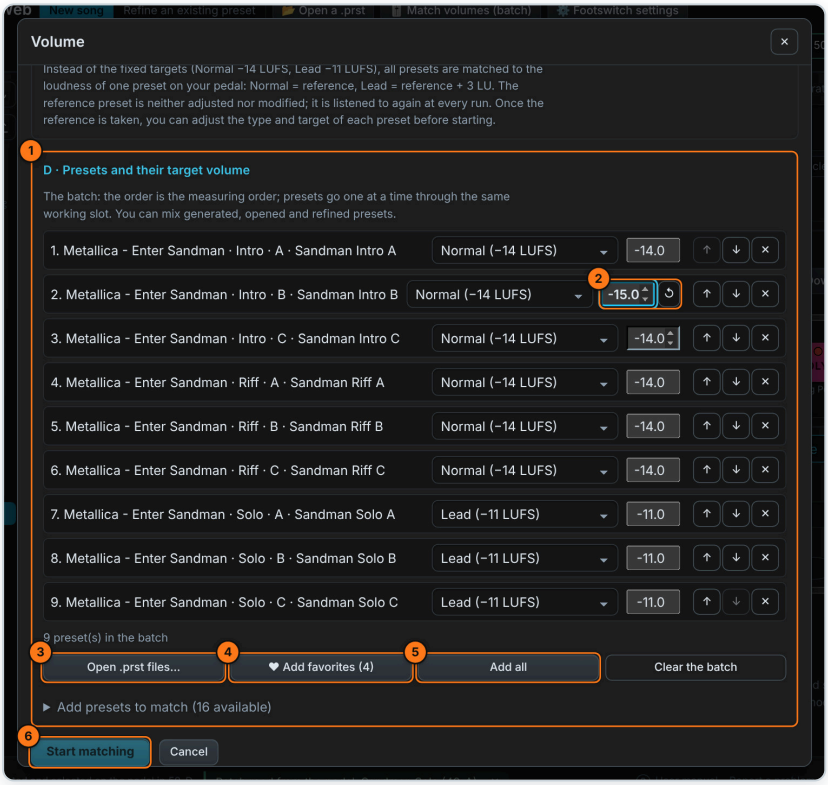


Figure 21 — The Volume window (open it with "Match volumes (batch)" or the Volume tab).

The window reads top to bottom: **A** the pedal, **B** the audio input (choose "GP-200"), **C** the working slot, **D** the presets and their target volume.

1. **The batch** (step D): the list of everything that will be matched, in measurement order. You can mix generated, opened or refined presets. ↑ ↓ reorder, × removes.
2. **The preset's target** (step D): a field with small ▲▼ arrows (they appear on hover or when you click in the field) for each preset. Each press changes the value by 0.5 LUFS. Here, the second preset has been changed: the field is highlighted and a ↺ button brings back the default.

3. **Open .prst files...**: adds several files at once. The window stays open and *nothing is sent to the pedal before you start*.
4. ♥ **Add favorites (6)**: puts all your favorites into the batch at once. A huge time saver before a gig.
5. **Add all**: the whole library. The "Add presets to match" panel lets you add them one by one.
6. **Start matching** (greyed out while the pedal or the audio input is missing).

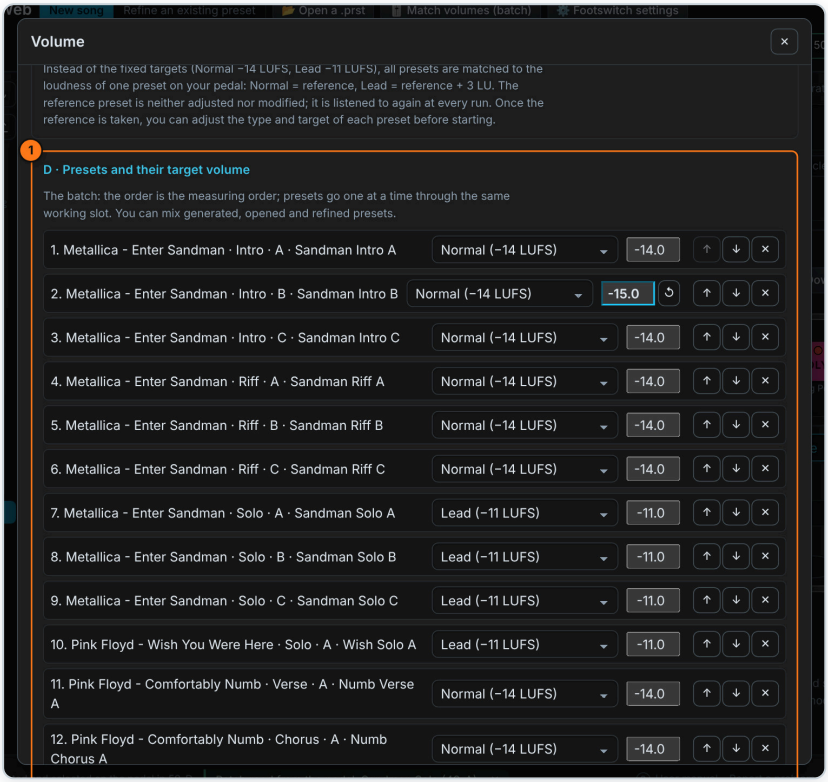


Figure 22 — After "♥ Add favorites".

For each preset, choose the **type**: **Normal (−14 LUFS on GP-200, −16 on GP-150/GP-180)** for rhythm parts and clean sounds, **Lead (−11 LUFS)** for solos, which must sit above the mix (on GP-200, 3 dB apart). All presets go, one at a time, through the same working slot (the slot chosen at step C).

Adjusting a preset's target. To the right of the type, each preset shows its **target in LUFS** (default −14 for Normal, −11 for Lead on GP-200; −16 and −11 on GP-150/GP-180) with small arrows (figure 21): each press changes the value by 0.5. Raising the value matches this preset **louder**; lowering it matches it **quieter**. The other presets do not change. The value belongs to each preset, is highlighted when it differs from the default, and a small ↺ button brings back the default. To avoid excess, the target can only move 3 LU away from the default: **on GP-200 −17 to −11 for Normal, −14 to −8 for Lead** (on GP-150/GP-180: −19 to −13 for Normal). If you change a preset's type, its target goes back to the default of the new type. Nothing is saved: the next time you open the page, everything starts from the defaults again.

> **Use sparingly**: −14 and −11 (−16 and −11 on GP-150/GP-180) are common reference points so that all your presets sound equally loud. A small offset (for example −13 for a slightly more present rhythm part, −12 for a solo that sits back a little) is almost always enough; the further the target moves from the default, the louder or quieter that preset will be perceived compared with the others.

Matching every preset to a reference preset (GP-200). In the *Reference preset (optional)* block, tick *Match all presets to a reference preset*: instead of the fixed targets (−16 / −11), every preset in the batch is matched to the level of a preset you like. There are two ways to give the reference:

- **Listen to a preset on the pedal**: pick it from the list, which shows the slot and the preset name (for example "02-C Zombie Clean"; any preset in the GP-200 memory, except the working slot, which gets overwritten and

appears greyed out). The names come from the “Pedal memory” column: until it has been read, only the slot number is shown. When you start, the pedal selects that preset (nothing is written to it), the app measures it like a preset of the batch — play your chord on a loop — and then matches the batch. The reference is measured at **every run** (about 30 seconds: it is accepted as soon as the uncertainty is ±0.8 dB or less) and is never kept from one session to the next. Once it is taken, the app stops on a **“✔ Reference taken”** banner (measured level, Normal and Lead targets): **click “OK, match the presets”** when you are ready to play your chord again. Measuring the first preset only starts after that click.

- **Type the value**: type a level in LUFS (from −30 to −6) if you already measured it; the pedal is then not used for the reference.

The target of a **Normal** preset is then the reference, that of a **Lead** preset the reference **+ 3 LU** (3 dB apart). In the form, per-preset custom targets are greyed out in this mode; however, **as soon as the “Reference taken” banner appears (and then during the whole measurement)**, each “to do” line of the list offers its **type** (Normal / Lead) and its **target**: you can raise or lower a preset (for example −18 for a solo that should stand out a little more), change its type, or go back to the calculated value with the ↺ arrow. These settings only apply to the current batch; nothing is written to the pedal until you click OK. The measured reference is shown above the list and appears in the measurement trace. The reference preset itself is never modified.

Very loud reference. If the measured reference is above about **−10 LUFS** (a typical level is more like −14), the “Reference taken” banner shows a warning: some presets may not be able to reach such a high level, even with every volume at maximum. Play more softly or lower the reference preset’s volume, then start the harmonisation again; you can also set each preset’s target before continuing.

> **Hardware.** Selecting the reference preset (Bank Select + Program Change) and measuring it were verified with a fake pedal and a fake microphone; they remain to be confirmed on your pedal. If the reference measurement does not become reliable, the app applies the same relaxation rules as for an ordinary preset.

> **The working slot is overwritten** during the whole operation: the app asks for confirmation (“Overwrite and start”). Choose an empty slot.

During the measurement

Adjusting targets on the fly (GP-200). In the preset list, every line that has **not been measured yet** (“to do”) offers its **type** (Normal / Lead) and its **target**: you can change them at any time, even while the app is measuring another preset, with the same rules as before starting (default value, limits, highlight, ↺ button, nothing is remembered). The preset being measured is locked. Changing a target sends nothing to the pedal and does not disturb the current measurement: the preset is measured with its new target when its turn comes.

Redo this preset (GP-200). A preset that is already validated also shows its type and target, with a **Redo this preset** button. You can first change the type or the target (this is only a draft: the validated result does not change until you click), then click: the preset is **redone right away**, starting from the **original file** and with the values shown. If another preset was being measured and was not validated, it goes back to “to do” and the app picks it up again as soon as the redone preset is validated (then carries on as usual). The new result replaces the old one in the list and in the ZIP. The reference is not measured again.

Once the harmonisation is **finished**, every line of the summary keeps its **Redo this preset** button: the app reminds you that the working slot will be overwritten again, then measures only that preset; the others keep their result. At the end, **Finish** takes you back to the updated summary.

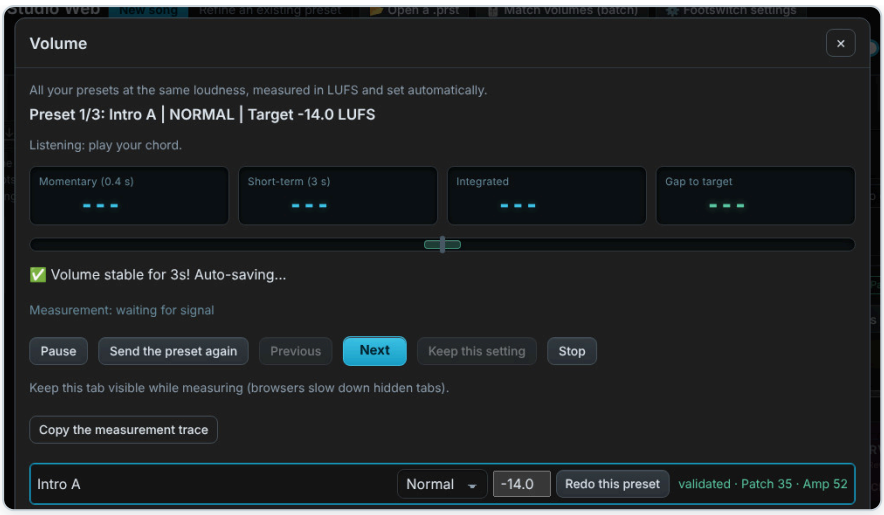


Figure 23 — Measurement in progress: momentary, short-term, integrated, gap to the target.

Play a chord on a loop, steadily, without changing your strength. On the GP-200, the app first adjusts the patch volume (down to 25 at the lowest), then the cab volume, then the amp volume as a last resort (it keeps the tone best), and gain only if the amp has no volume; anything that has drifted from its original value in the opposite direction goes back toward it first. On the GP-150/GP-180 the historic order is kept (amp, cab, patch, gain). As soon as the level is within tolerance (0.4 LU) for **3 stable seconds** *and the measurement is reliable* (see below), the preset is validated and saved, then it moves on to the next one, with no click between two measurements.

Buttons: *Pause*, *Send the preset again*, *Previous*, *Next* (*Finish* for the last one), *Keep this setting* (validate as is), *Stop*.

The reliable measurement

When you play chords, the volume keeps changing (attack, decay, pick strength): a measurement taken over a few seconds is bound to be approximate. Under the status message, one line tells how far the app can trust what it hears: *"Measuring: 8 s of playing kept · uncertainty ±0.9 dB"*, then, in green, *"Reliable measurement: ±0.3 dB"*. The app:

- only adjusts the volume if the gap to the target is **larger than the uncertainty**: otherwise it waits and keeps listening, instead of correcting noise and then correcting its own correction;
- only validates a preset once the measurement is reliable (at least three 2-second batches of playing, and an uncertainty of ±0.5 dB at most);
- starts a fresh measurement after every adjustment, without keeping the old gap. On the GP-200 it also **discards the first half second** of audio that follows an adjustment: that little bit still has the old level and, without this, it skewed the average for 20 to 25 seconds.

> **To be fast and accurate**: always play the same chord progression, with the same attack, in a loop for 10 to 15 seconds per preset. The steadier your playing, the faster the uncertainty falls. If your playing varies a lot from one chord to the next, the measurement takes longer; after 45 seconds without a reliable enough measurement the app relaxes its requirements rather than stay stuck (and *Keep this setting* is always available).

The app learns how your preset responds. For the **amp**, the **cab** and, on the GP-200, the **patch volume**, it starts from a simple law: their volume acts like a gain, so going from a value A to a value B changes the level by about 20 × log10(B ÷ A) LU (example: a cab going from 100 to 78 drops the level by about 2.2 LU; a patch volume going from 50 to 28 drops it by about 5 LU, as checked on a real GP-200). From the very first adjustment it therefore works out the value that brings the level back to the target, instead of a fixed step per step: one step matters less

when the value is high than when it is low. For the gain (and, on the GP-150 / GP-180, the patch volume) it starts from 0.25 LU per step. After each adjustment it compares the change it actually measured with the expected change, works out the real response of that module (amp, cab, patch volume or gain) as a correction coefficient around the law (amp, cab, and patch on the GP-200) or a value per step (gain), and uses it to size the next adjustment: fewer back-and-forth corrections when a preset reacts much more or less than expected. It only learns when the adjustment is large enough compared with the measurement uncertainty; a doubtful value is ignored, and the retained response is always bounded (coefficient from 0.4 to 3 for the amp, the cab and the GP-200 patch, 0.12 to 0.50 LU per step for the gain). Everything stays in memory for **the current preset**: nothing is saved and nothing carries over from one preset or session to the next, so each user adapts to their own presets and their own playing.

A window that stays readable. During the measurement, everything you need to follow it (meters, bar, message, buttons, trace copy) stays displayed at the top; only the **preset list** scrolls below it, and follows the current preset. Even with dozens of presets, the live measurement never leaves the screen (on a small screen the display is unchanged).

Copy the measurement trace (during the measurement: above the list; after the matching: below the list) copies the details of the operation: one line per second and, for every adjustment, the expected effect then the effect actually measured, with the learned response (in LU per step) at each stage. Nothing is sent: paste it in a message if you want to understand why a preset took long to set.

The result

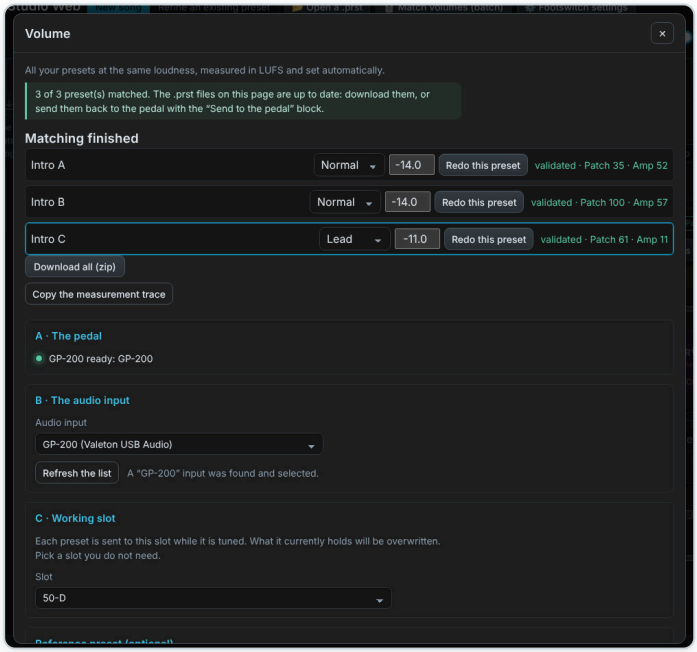


Figure 24 — End of the matching: values kept for each preset.

- The page's `.prst` files are **updated** (✓ in the library). They are not in the pedal yet: send them with the *Pedal* tab or the *Set list*.
- **Download all (zip)**: one `.prst` and one `.json` per preset in the batch.

> **Hardware.** The measurement (LUFS) and the adjustment rules are those of the desktop program, verified with a fake pedal and a fake microphone. The web app is more careful on one point: it takes the reliability of the measurement into account before adjusting or validating (see above). That point, the learning of the preset's response and the custom targets were verified by simulating chord playing and with a fake pedal, not yet on a real pedal. On GP-150 / GP-180, the measurement constants are taken from the GP-200: every write is verified by reading back.

12 CTRL: multi-effect footswitches

CTRL 1 to 8 are the preset's "switches": each one turns on or off, with a single press, the modules you tick.

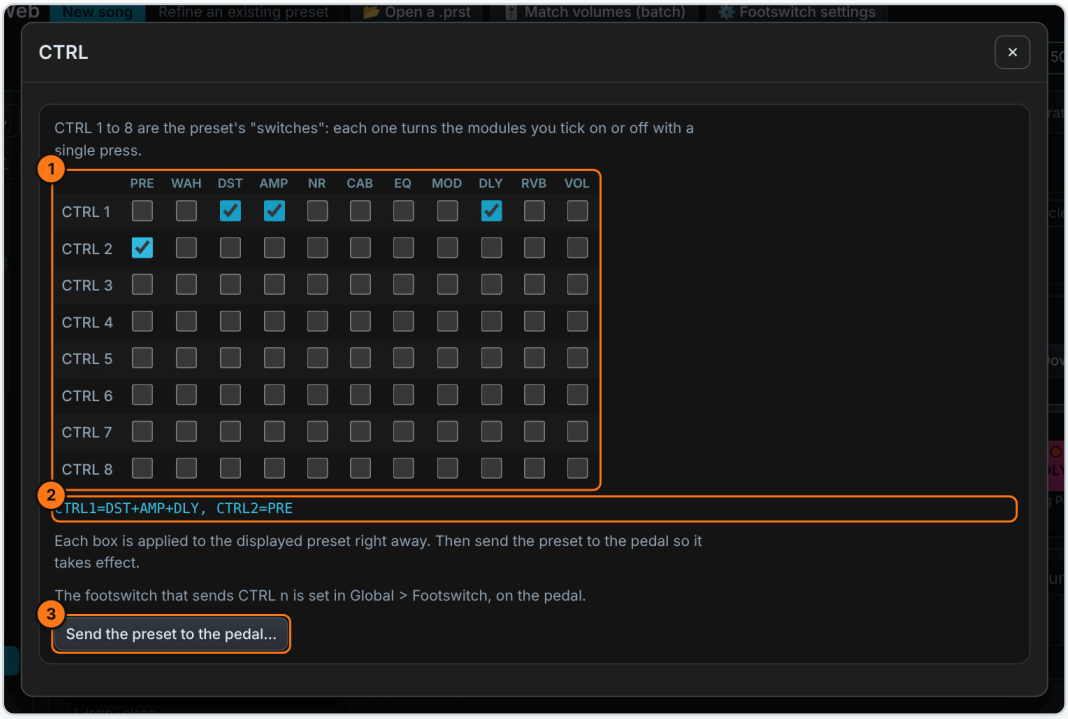


Figure 25 — The *CTRL* tab: an 8 × 11 grid.

1. **The grid**: one row per CTRL, one column per module (PRE, WAH, DST, AMP, NR, CAB, EQ, MOD, DLY, RVB, VOL). Each ticked box is **applied immediately** to the displayed preset.
2. **The summary** in plain words, here `CTRL1=AMP+DLY, CTRL2=PRE`.
3. **Send the preset to the pedal...**: so that the pedal takes it into account.

> **Which footswitch sends which CTRL?** This is set with the ⚙ *Footswitch settings* button (below), or on the pedal, in *Global* > *Footswitch*.

The CTRL tab and the *Footswitch settings* button do not exist on GP-150 / GP-180.

The "⚙ Footswitch settings" button: the function of each footswitch (GP-200)

This button appears in the top bar, next to *Match volumes (batch)*, when a GP-200 is connected. It opens a wide window where everything is visible at once. It reads the function of each footswitch from the pedal, lets you change it, and sends it to the pedal.

1. **The three modes**: *Patch*, *Stomp* and *User*. The ★ marks the one the pedal is using right now.
2. **The table** (left): for each of FS1 to FS8, a function on **TAP** (short press) and one on **HOLD** (press and hold). On the right: four **combinations** (FS1+5, FS5+6, FS6+7, FS7+8: two footswitches pressed together). The lists only offer what the pedal accepts: 26 functions for TAP and combinations (*None*, CTRL 1 to 8, Looper, DRUM, Tuner, TAP, Bank +/−, Patch +/−, Patch A to D, Bank, Drum Patch +/−, EXP1 A/B, MIDI), 17 for HOLD (no CTRL or TAP).

3. **Every change is sent to the pedal right away.** The screen shows the change **only after the pedal confirms it**: the pedal sends the frame back, then the page reads its configuration again, and that read-back is what is displayed. If the pedal does not confirm, the screen stays as it was and a message says so.
4. **The pedal's active mode** and **FS Mode** (*Wait* / *Initial*) are set under the combinations, on the right. Changing the active mode acts on the pedal immediately; the page never does it by itself.
5. **Reset the … mode to factory** replaces the 20 functions of the displayed mode with the factory values, after a confirmation.

> **Going back.** Before the very first change, the page downloads **a backup** of the configuration read when the tab was opened (`footswitches_gp200_YYYYMMDD_HHMM.json`). After that: **Revert to the original** puts that configuration back (the list of changes is shown before sending); **Download a backup** saves the current configuration; **Restore from a file…** reads a backup and sends only the differences, after confirmation.

The following are **not** covered: the expression pedal, external pedals (EXT FS), the global EQ, general settings and MIDI.

13 GP-150 and GP-180

The page recognizes the connected pedal and switches to the matching profile. The **GP-200 / GP-150 / GP-180** selector is at the top of "My presets"; each model has its own library.

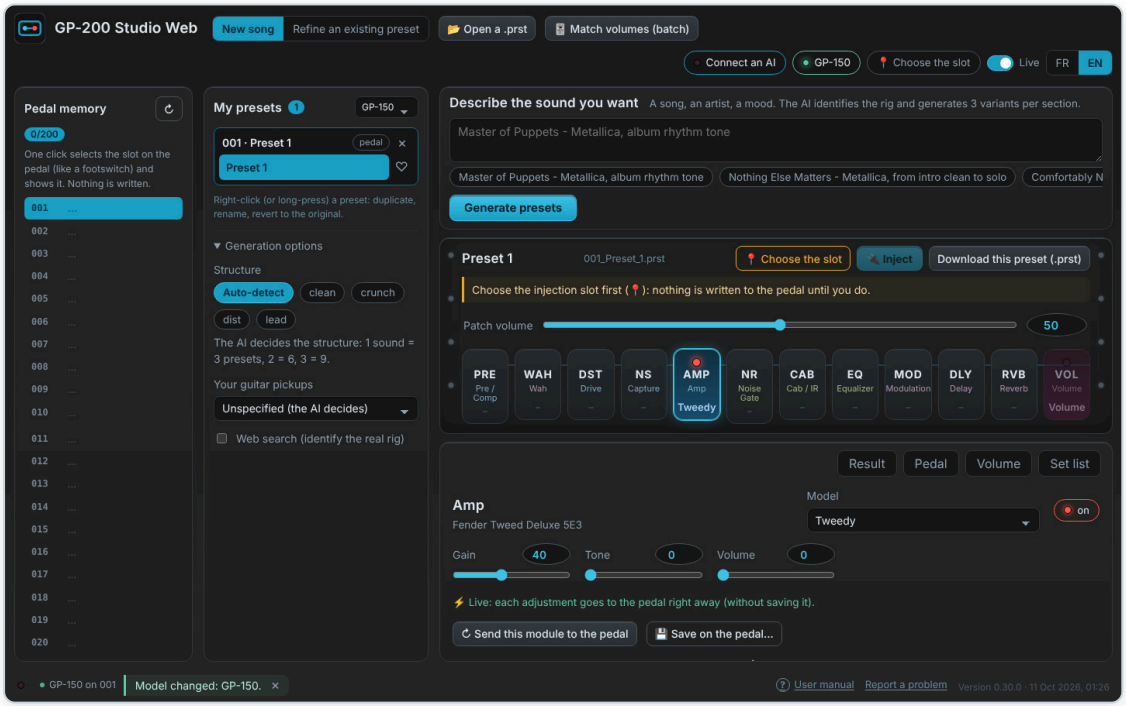


Figure 26 — GP-150 profile: 200 slots numbered 001 to 200.

	GP-200	GP-150 / GP-180
Slots	256 (01-A to 64-D)	200 (001 to 200)
Modules	11 (PRE, WAH, DST, AMP, NR, CAB, EQ, MOD, DLY, RVB, VOL)	12 (PRE, WAH, DST, NS, AMP, NR, CAB, EQ, MOD, DLY, RVB, VOL)

Chain order	fixed	free
Preset name	16 characters	13 characters
.prst file	1,224 bytes	1,128 bytes
CTRL tab	yes	no
Rearrange the memory (drag and drop, Restore / Import / Export icons, right-click menu)	yes	no (column unchanged)

Reading, memory of the 200 slots, live editing, listening, sending (injection, set list), AI (generation and refinement) and volume matching work as on GP-200, with a **verification read after every write**. The “NS” models (SnapTone / A2 Lite captures) and user IRs are not offered to the AI, but are kept as they are when refining. The app never sends a factory reset.

If the MIDI port is not recognized with certainty, a message asks you to confirm the model (*Use this model* button) before any write.

> **Not tested on hardware.** No GP-150 / GP-180 pedal was available during development: all writes, GP-180 reading, 22 effect IDs outside the catalog and the LUFS constants were verified with a simulated pedal only. The interface labels say so. Data and protocol come from the open-source project *valeton-gp150-editor* (MIT license).

14 Custom firmware (Goaltoday)

This chapter only applies to GP-200 pedals running Goaltoday's **custom firmware**. If your pedal has the original firmware, skip to the next chapter: **nothing described here exists in Standard mode**.

> **Standard or Custom?** Every other chapter of this manual describes **Standard mode**, the normal mode of the app and the one it always starts in. **Custom** mode is an option you choose yourself with the *Firmware* switch (see below); as long as you do not turn it on, nothing in this chapter applies and nothing special is ever sent to your pedal.

What the custom firmware adds

Goaltoday's custom firmware is a free project (GPL-3.0 licence) that replaces the GP-200's original firmware. It adds **"universal"** blocks (any model in any block), **NAM** models, four **scenes** per preset (A to D) and a **Dual Chain** (two effect branches played in parallel). GP-200 Studio can open, display and **edit** these presets, then write them to the pedal.

1. Switching to Custom mode

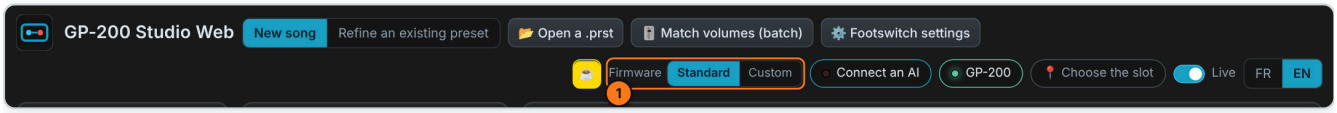


Figure C1 — The Firmware switch, here on Standard (the default).

1. **Firmware: Standard / Custom**, in the top bar (GP-200 only). It starts on **Standard**. When you connect a pedal, the app asks it one read-only question: if it runs the custom firmware, the app **switches to Custom by itself** and tells you; otherwise it stays on Standard without a word. If you switch back to Standard yourself, the app respects your choice. You can also click **Custom** by hand. It is visible even with no pedal connected: you can then prepare presets offline.

If you switch to Custom while your pedal has the original firmware, it does not answer custom requests: a message tells you so and nothing else is attempted. Simply go back to Standard. Conversely, in Standard mode a preset that contains custom-firmware blocks shows a banner and sending it to the pedal is blocked (an original pedal would not know how to read it).

2. Opening a file and understanding the principle

In Custom mode you can open a **.gpscene** (preset + scenes, the format of Goaltoday's editor), a plain **.prst**, or read the pedal's preset (⋯ menu, *Read from the pedal*). The principle is the same as in Standard mode: the preset shown is **written by itself to your working slot** and each change follows after about 2 seconds (see chapter 6). With no pedal, or if you untick *Inject automatically*, you edit the file on screen and then write it by hand.



Figure C2 — A file open in Custom mode (here with no pedal connected).

1. **The action bar**: **‖ Dual Chain**, A / B blend and offset, **Apply to the file** and **Cancel** (these two buttons only appear without automatic writing: with no pedal, for example). An edit stays on screen (the setting changes colour) until you click **Apply to the file**; **Cancel** goes back to the file's state.
2. **The rack**: the block chain, with the patch volume above it. In Custom mode the chain itself is the Dual Chain editor.
3. **Write to the pedal**: with automatic writing it only appears after a failed write (to try again); otherwise it is the only button that needs the pedal (greyed out without a pedal, and while some edits are not applied yet). As soon as the file has been modified (new model, or "Apply to the file"), it is **highlighted**: the sign that the file still has to be written to the pedal to hear the change.
4. **⋯**: **Export with scenes (.gpscene)**, **Read from the pedal**, **Scenes from a capture…** and **Custom firmware details** (pedal capabilities, Dual Chain, scenes, installed NAM models).

Everything is adjusted as in Standard: click a block to open its card, change its settings, switch a block on or off with its **LED**, set the **patch volume**. In Custom mode these gestures change the **file**, with or without a pedal connected, and the writing to the pedal then follows by itself. Only a block's **LED** in a preset without scenes acts on the pedal right away: you hear it before the writing has even finished. A common setting is copied everywhere in the file (base preset, active preset and scenes).

3. Choosing the model of any block



Figure C3 — An amp in the WAH block.

1. **The block** takes the symbol, colour and name of the family of the model it holds when it differs from its slot, with the original slot shown small: "AMP ▸DLY" = an amp in the DLY block.
2. **The Model list** offers the models of **all families**, grouped (Amps, Cabs, Distortion, Delays, NAM…); the block's own family comes first.
3. **The LED**: a block that is **off** and receives a model from **another family** is switched on automatically (otherwise its branch would stay silent).

The choice **changes the file right away** (base preset, active preset and scenes); the settings start again from the new model's defaults. Pending edits (Dual Chain, scenes, settings) are applied to the file first. To go back, reopen the original file. With the Dual Chain this allows, for example, **two amps in parallel**, one per branch, each with its own cab.

4. Dual Chain: two branches in parallel

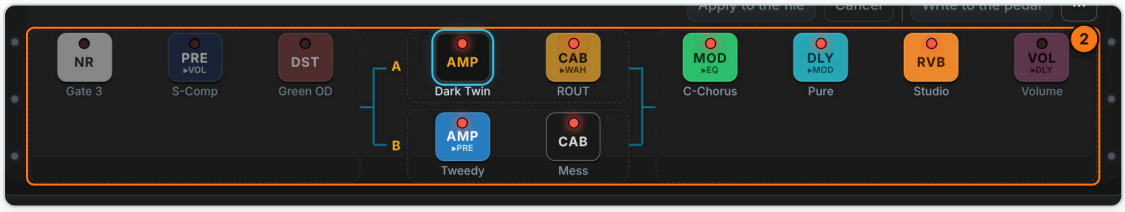


Figure C4 — Dual Chain: before zone, branches A and B, after zone.

1. **‖ Dual Chain**: turns the Dual Chain on or off (right-click on the chain does the same; the button is handier with a finger). On a simple chain it opens an **empty** parallel section (A and B empty) at the preset's *Return* position: no block moves and nothing is sent to the pedal until you drop a first block into A or B. To open the section **right after a given block**, **right-click that block** and choose the parallel option: with the chain NR · PRE · WAH · DST · DLY…, a right-click on DST opens the empty A / B section right after DST. Turning it off gives the simple chain back.
2. **The chain**: with Dual Chain off, **drag the blocks** with the mouse to change their order, exactly as in Standard mode (the order is written to the pedal by itself). With Dual Chain on, it splits into **Before**, then **A** above **B**, then **After**. The blocks of both branches are played at the same time, then mixed. **Drag the blocks** from one zone to another (with the mouse; with a finger, long press then drag; Esc cancels). Another way: right-click a block, *Move to…*. The right-click also brings the block **to the front** (its panel is shown) without turning it on or off. **Several blocks at once**: click a block, then **Shift+click** (or Ctrl / ⌘+click) another one in the same

zone: every block between them is selected (highlighted in blue); drag one of them to move the whole group, in the same order. Esc or a click elsewhere clears the selection (mouse only, not on touch yet).

3. **A / B blend**: A only on the left, B only on the right, 50 % in the middle.
4. **A / B offset**: −32 to +32 samples ("+" delays A, "−" delays B).
5. **Phase B**: inverts the phase of branch B.

Without automatic writing, nothing changes in the file until you click **Apply to the file** (with it, this happens by itself after about 2 seconds). Applying changes the open preset **and all its scenes**. To turn the Dual Chain off: the ‖ button, then Apply; its settings stay stored in the preset. The Dual Chain reuses the preset's FX Loop Send / Return positions.

When the window is wide enough, the scene bar (see 5) and the action bar (see 1) sit **on the same row**; on a narrow screen they stack.

5. Scenes A to D

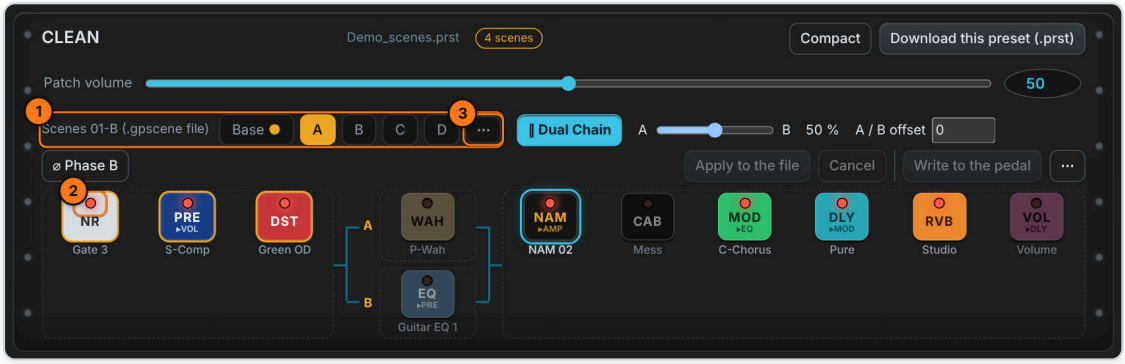


Figure C5 — A preset with 4 scenes, scene A shown.

1. **The scene bar**: **Base · A · B · C · D** (a dot marks the scene active on the pedal). Clicking a scene shows on the rack the blocks that are on in that scene; those that differ from the Base are outlined in orange. A click on **A, B, C or D** also asks the pedal to switch to that scene: the editor then reads the active scene back to check. If the pedal does not confirm, the scene is only displayed, a sentence says so and there are no more attempts until the next launch. The **Base**, and the scenes of an opened file (.gpscene), stay display-only. When you change scene **with your foot**, the bar and the rack follow: the dot moves to the new scene and the rack shows its blocks.
2. **A block's LED** switches it on or off **in the scene shown** (in the Base view, it is the preset itself).
3. **⋯**: **Copy Base here**, **Copy scene X here**, **Restore this scene**.

The parameters that scenes can vary are marked ◆ in the block card; they are edited while a scene A to D is shown (in the Base view you change the Base value; each scene keeps its own). If the preset **has no scenes**, the **Enable scenes** button creates the four scenes by copying the preset's current state. A .prst opened on its own has no scenes (they are not stored in a .prst).

Reopening a preset with its scenes. Choose (or drop) the **.prst**, the **.gpscene** and the **.json** of the same name together: the app opens a single preset, with its scenes and their names (the .json is optional). If the files are not the same preset, the .prst opens alone and a message explains why the scenes or names were refused. For a preset already open, the **⋯ menu** offers **Scenes from a .gpscene…** and **Scene names from a .json…** (nothing is changed if it is not the same preset).

A single file: the ZIP. ⋯ → Export as ZIP (preset + scenes + names) creates a ZIP with the preset's .prst, .gpscene and .json (a generation's "Download all" has the same shape). Just reopen it: the app brings back preset, scenes and names. Only ZIPs that contain a .gpscene are opened this way. The "Export with scenes (.gpscene)" entry is still there.

6. Writing to the pedal (automatic)

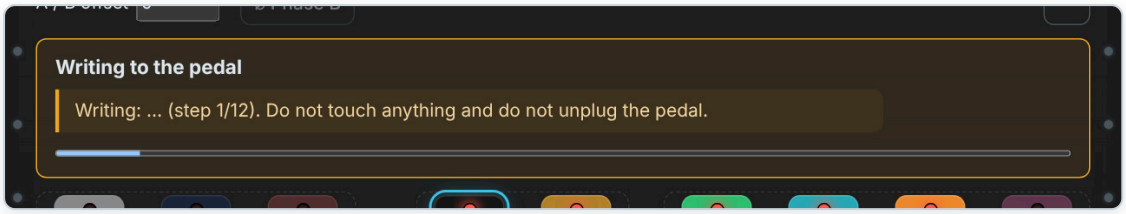


Figure C6 — The progress shown during an automatic write.

There is nothing to click. As in Standard mode, the app works in your **working slot** (the one you confirm in the 📍 window; the *Inject automatically* box must stay ticked):

- **You open or choose a preset**: it is written to the working slot by itself, with its scenes if it has any, and the pedal **stays on that slot**. A preset you have just read from the pedal is not written again until you modify it.
- **You change something** (Dual Chain, scene, setting, model…): after about **2 seconds of calm**, the change is applied to the file and written to the pedal. Several settings in a row go out in a single write. The **Apply to the file**, **Cancel** and **Write to the pedal** buttons disappear.
- **A small green message** at the bottom confirms the write is done; there is no window to close.

> **Backups.** Before the **first** overwrite of a slot, the app reads back that slot and the active preset and **downloads a ZIP**: that is your original. After that, no ZIP at each write, nor in a later session as long as the slot still holds what the app wrote there; the following writes are also faster. When a setting sounds good, ⋯ → **Export with scenes (.gpscene)** or **Export as ZIP** keeps the current state, at any time.

> **If a step does not go as expected**, everything stops without re-sending the command involved (only the original footswitch mode is put back): the message names the step, and a **ZIP report** and a **log** can be downloaded. The app does not retry by itself: the **Write to the pedal** button comes back so you can run it again by hand.

Do not unplug the pedal during the write. A **single preset** (a .prst, or a preset without scenes) is written the same way, without scene restoration: the pedal builds the slot's own scenes. If you untick *Inject automatically*, or as long as the working slot is not confirmed, nothing goes out by itself: you get **Apply to the file** and **Write to the pedal** back (which writes to the working slot, or to the pedal's current preset if there is none).

> **Option: do not save to flash.** In the ⋯ menu, *Do not save to flash (trial)* sends every change to the pedal (you hear it) without saving it, as in Standard. The **Save to the pedal** button then appears: it saves the preset already sent. If you change preset without saving, the change is lost. Off by default.

> **Option: live settings.** In the ⋯ menu, *Live settings (trial)* sends the setting you move (and the LED) to the pedal right away, as in Standard, without rewriting the preset: only that setting is sent. It applies to a preset without scenes: block settings (universal blocks included), LEDs and **changing the model of a block** are sent on their own, without rewriting the preset (the pedal is read back to check; if it did not take the change, the change is written as usual). Other changes (Dual Chain, block order, scenes) are written as usual. The **Save to the pedal** button appears to keep the result; if you change preset without saving, the change is lost. On by default; untick it in the ⋯ menu to have every change written to the pedal as before.

> **Moving a block or changing the Dual Chain.** When it is the only pending change, the chain is sent to the pedal (you hear it) but **is not saved to flash**, as in Standard. The message says "not saved" and the **Save to the pedal** button appears: click it to keep the change. If you change preset without saving, the move is lost.

Tweaking a preset from the pedal (for example 30-C). When the displayed preset was read from the pedal and the pedal is still on that slot, your changes are written **to that slot** (30-C), not to the working slot, as in Standard where you tweak the current patch. Nothing is saved to flash by itself: the message says "not saved" and the **Save to the pedal** button writes the result into 30-C. Before the first write to that slot, a backup ZIP of the original is downloaded. The working slot stays reserved for files (opened or generated) and for volume harmonization. If you change patch on the pedal without saving, the change is lost.

Warning. The working slot is overwritten at each write. Choose a free slot, or a slot whose ZIP backup you have kept.

7. Generating a preset with AI in Custom mode

In Custom mode, the *Generate presets* button uses a **dedicated prompt**, separate from the Standard one: the AI knows what the custom firmware can do. Standard mode keeps exactly its usual prompt. You describe the sound as usual; in addition, the AI can:

- **Put a model in any free block**: an amp or a cab in an effect block, a second delay or reverb in a free block, and so on. Two effects one after the other (for instance two delays for a U2-style sound) are simply placed in series, no Dual Chain needed.
- **Build a Dual Chain** when the sound calls for it: two parallel branches (amp + reverb, two amps, dry signal + effect), with blend, offset and phase.
- **Build scenes**: when a song's variations boil down to switching blocks on / off or changing a few values (a more saturated chorus, a solo with delay), the AI writes them as scenes A to D rather than as separate sections. A preset can thus carry up to five sounds: the Base and four scenes. The three variants of each section each carry these scenes. The Base is the main sound (verse, intro): **scene A is always a copy of it**, so you can get back to it with one footswitch press, and scenes B, C and D carry the real variations (D is optional).

The AI gives **each scene a name** ("Chorus", "Solo"…): it appears in the tooltip of the A to D buttons and next to the scene bar. A scene B, C or D **identical to the Base** (so unused) is shown greyed out; it can still be selected and edited. Scene A, identical to the Base by design, keeps its normal colour. Names are not saved in the .gpscene file: they are kept during the session and in the .json file of the ZIP archive.

Every reply is checked by the app before being accepted: an unknown model, an impossible block, or scenes too heavy to fit in the preset's memory are sent back to the AI, which corrects them. If nothing valid comes out, you get a clear message and nothing is added to the list.

The resulting presets open like any other Custom-mode file: you can edit them, then **write them to the pedal** (see above). A preset that contains scenes downloads as a **.gpscene** (preset + scenes); without scenes, it is a **.prst**. In the ZIP of generated presets, each variant with scenes is provided as a .gpscene.

The 🎤 *Optimize for live* button and preset refinement remain those of Standard mode.

8. Reporting a problem in Custom mode

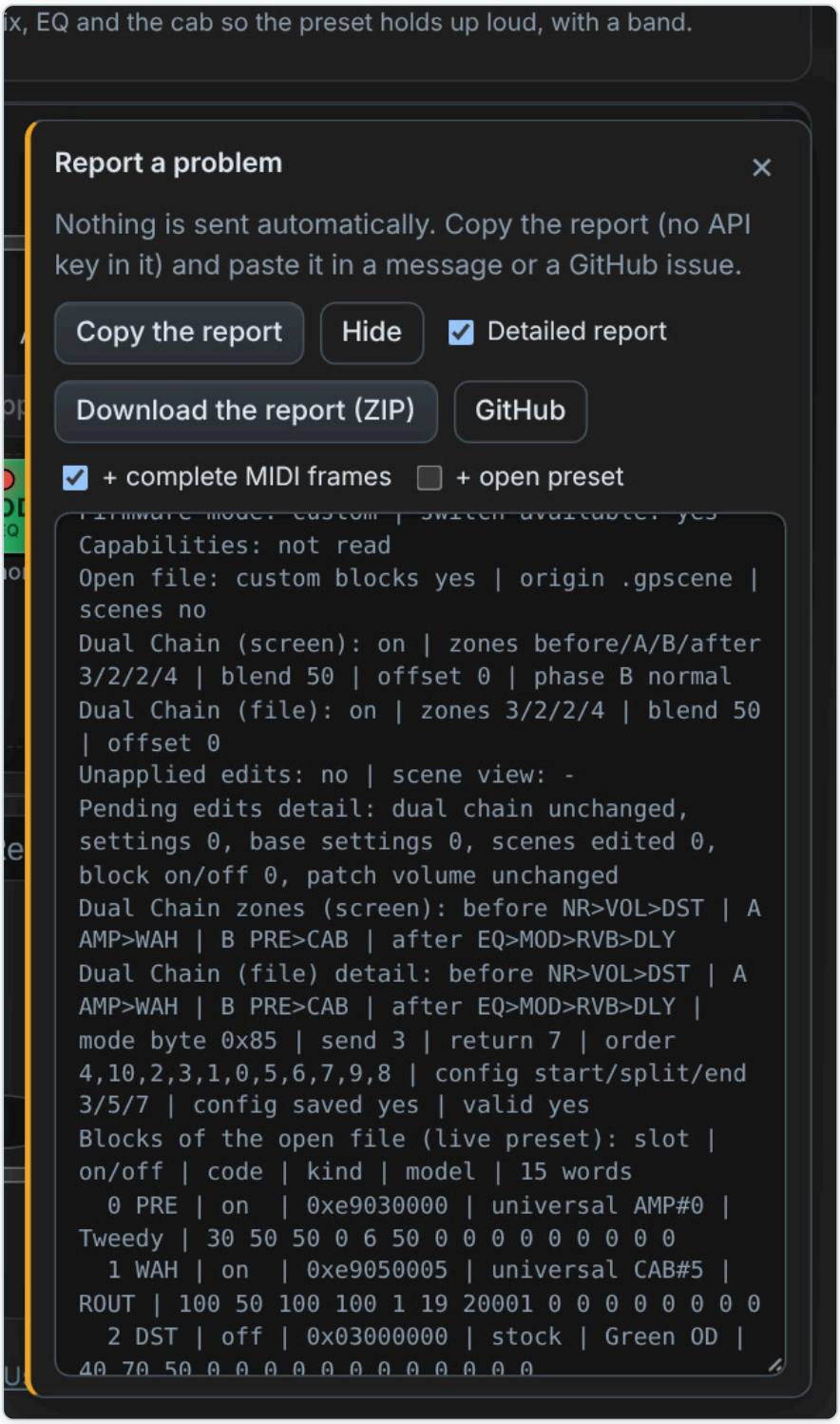


Figure C7 — The detailed report (box ticked by default in Custom mode).

The **Report a problem** link (at the bottom of the page) gathers, in two clicks, everything needed to understand what you were doing: click **Copy the report** and paste it into a message, optionally adding a line about what you wanted to do. In Custom mode the detailed report contains:

- the custom firmware **mode** and state (capabilities read from the pedal, open file);
- the **list of blocks** of the file: position, on / off, **model code and name** of each block;
- the **Dual Chain** in detail (zones, blend, offset), the patch volume and the scenes;
- the **log of your edits** in the session (model change, *Apply*, *Cancel*) with their result;

- your **latest on-screen actions**, the **latest MIDI frames** sent and received, and the **last write** step by step;
- the **last AI generation** in Custom mode: sections, scenes, Dual Chain and the AI's rejected replies (without the text of your request).

Nothing is sent automatically. The report contains **no API key, no file name and no preset name**. It disappears when you close the page.

9. Good to know

- Live editing, loading NAM models and applying the state of an active scene from the file are not supported yet. In AI generation, two amps switched by scenes are not offered (use Dual Chain).
- Custom mode changes nothing for Standard mode: going back to Standard puts the app back exactly as described in the other chapters.
- The custom firmware is Goaltoday's project, independent of GP-200 Studio and of Valeton.

15 On phone and tablet

Below 1100 px of width, the page stacks: request area, preset list, rack, module sheet.

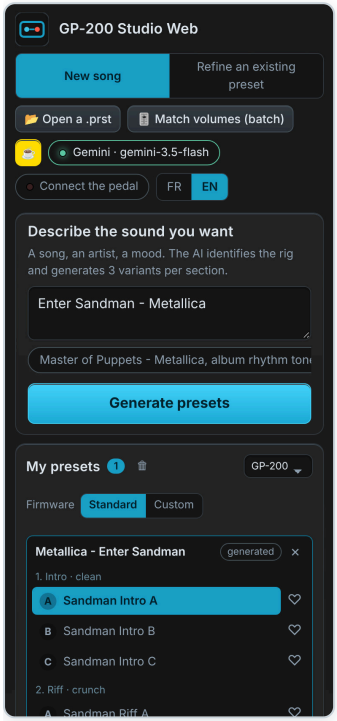


Figure 27 — Mobile layout.

- Large touch targets are kept (main button 48 px).
- **Long press** on a preset = right-click menu.
- Drag and drop in the memory is not supported on touch: select a slot then use `Alt` + arrows with a keyboard, or do the rearranging from a computer.
- To send presets from Android, plug the pedal in over USB OTG (chapter 8).
- iPhone and iPad: no MIDI in the browser, but you can generate, open and download presets.

16 Reporting a problem

A **Report a problem** link at the bottom of the page opens a diagnostic panel. It also opens by itself at the first unexpected error.

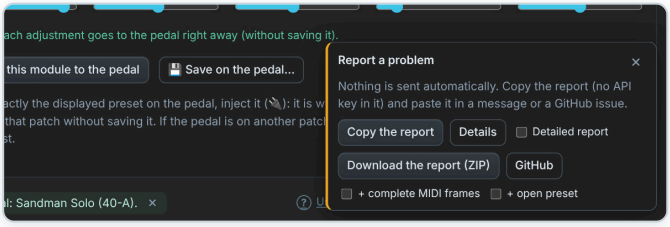


Figure 28 — The “Report a problem” panel.

- **Nothing is sent automatically.** *Copy the report* puts it on your clipboard; paste it in a message or in a GitHub “issue” (*GitHub* button). *Details* shows the full text.
- The report contains: version, browser, AI provider and model, MIDI state (ports seen, port chosen, pedal patch) and the last 80 events; after a matching, it also adds the last 120 lines of the **measurement trace** (chapter 11).
- **Download the report (ZIP)**: builds a ZIP file on your device (nothing is sent), handy to attach to a message when the text is long. It always contains **rapport.txt** (the detailed report). Two checkboxes add **the complete**

MIDI frames (the text only shows their beginning; ticked by default in Custom mode, with the frames of the last write) and **the open preset** (with its scenes in Custom mode; unticked by default). These additions may contain preset names: untick them if you do not want to share them. The same button is also in the window shown when a Custom write fails.

- **Detailed report** (checkbox in the panel; ticked by default in Custom mode): adds the mode, the **last 40 actions** (clicks, drags, keys, settings, with the time) and the **latest MIDI frames** sent and received. Actions are described by the name of the button or area, never by typed text, a preset name or a file name. This log stays in the page and disappears when you close it. The extra content of Custom mode is described in chapter 14.
- **No API key** appears in it: keys (Google, Anthropic / OpenRouter `sk-…`, `Bearer`, `api_key=…`) are masked as `[key hidden]`. The address of a local page is masked too.
- The **Log** tab keeps the course of the generation.

17 Troubleshooting

Symptom	Likely cause	What to do
The pill shows "USB ✕"	Browser without Web MIDI (Safari, Firefox, iPhone)	Use Chrome, Edge or Opera. You can still create and download presets.
No MIDI permission prompt, or "MIDI refused"	Page opened outside https (local file, attachment), or permission denied	Open the page from its https address (or `localhost`). Allow MIDI again in the site settings.
The pedal is not detected	The Valeton editor (or another tab) holds the port	Close it, then replug the pedal and click the pill. On Android: check the OTG adapter.
The Valeton editor no longer sees the pedal	GP-200 Studio keeps the port open (listening to the pedal)	Untick "Listen to the pedal" in the 📍 panel, or close the app's tab.
Footswitches: "the pedal did not confirm"	The pedal did not answer the frame (port held by the Valeton editor, cable)	Nothing was changed. Close the Valeton editor, click *Read the pedal again*, then try again.
Nothing happens when I move a slider	*Live* is off, or the pedal is on another patch	Turn *Live* on. If a banner says "other patch", click *Send the preset again*.
A setting cannot be heard live	It carries the ↻ symbol (silent live)	Send the module again or write the preset: it will be taken into account on loading.
My settings vanished when I changed patch	Live settings are temporary	Write the preset into a slot (*Save on the pedal…*).
"The API key is refused by the provider"	Wrong, expired or out-of-quota key	Check the key with the provider and paste it again (AI pill).
Gemini: "overloaded" / error 503	Free key during a busy period	Try again a bit later; the app switches to *flash-lite* by itself.
After the matching, one preset sounds louder or quieter than the others	A custom target was set on that preset (field to the right of the type, highlighted when it differs from the default)	Go back to the default target with the ↻ button, or set the type (Normal / Lead) again: the target returns to the default. Then run the matching of that preset again.
The matching stays on "Measuring" or takes a long time	Your playing varies a lot from one chord to the next: the uncertainty falls slowly	Play the same chord progression, with the same attack, in a loop. After 45 s the app relaxes its requirements; you can also click *Keep this setting*.
No sound during the volume measurement	Audio input not chosen, or tab hidden	Step B: choose "GP-200". Keep the tab visible. Allow the microphone.
The memory stays at "…" or stops halfway	First read got no answer, or pedal busy (the page tries once again by itself)	If it does not restart, click ↻. Reading waits while a send is in progress.
"The content of 11-C changed since it was displayed"	A slot was modified since the read	Nothing was written. Read again with ↻ and redo the plan.

"... refused: a valid GP-200 .prst is required" or "... go past the last slot" when importing	File of the wrong size, invalid checksum, preset from another pedal (GP-150 / GP-180), or too many files for the remaining room	Nothing was imported. Export the preset again from GP-200 Studio, or choose a slot higher up the list.
"To check: 11-C" after a move	The verification read differs from what should have been written	Use **Restore...** with the backup ZIP, or redo the operation.
My presets disappeared	The page was closed or reloaded	Libraries do not survive closing: download (ZIP) before closing.
A preset name is truncated or has no accents	Pedal limit: 16 characters (13 on GP-150/180), ASCII	Normal: the name that will be sent is announced before sending.

If the problem persists: **Report a problem** at the bottom of the page, copy the report and attach it to your question.

18 Quick reference

Shortcuts and gestures

`←` `→`	Previous / next module in the rack
`Esc`	Closes the floating window (tab, slot, settings)
Right-click / long press / `Menu` / `Shift`+`F10`	Preset menu: duplicate, rename, revert to original, delete the copy
Drop a `.prst` on the page	Open one or several presets
Drag a slot (Memory column)	Move it (plan, nothing is written)
`Ctrl` / `⌘` + click · `Shift` + click	Multiple selection in the memory (add · range)
`Alt` + `↑` / `↓`	Shift the selection by one step in the memory
`↑` / `↓` in a preset's target field (Volume window)	Raise / lower the target by 0.5 LUFS (same as the field's small arrows)
Click on a selected module	Turns it on / off (the LED always does it with one click)

Slot numbering (GP-200)

64 banks × 4 letters = **256 slots**: 01-A, 01-B, 01-C, 01-D, 02-A... up to 64-D. On GP-150 / GP-180: 001 to 200.

The 11 GP-200 modules

PRE	Pre / compressor: OD, boosters, auto-swell, hold...	**MOD**	Modulation: chorus, phaser, flanger, vibrato...
WAH	Wah	**DLY**	Delay
DST	Distortion / drive	**RVB**	Reverb
AMP	Amp	**EQ**	Equalizer
NR	Noise reduction (noise gate)	**VOL**	Volume
CAB	Cab / IR		

Glossary

Preset, patch
A complete sound: the 11 modules, their settings, the volume and the CTRLs. “Patch” is the pedal’s word, “preset” the app’s; they mean the same thing.

.prst
The file of a preset, in the Valeton editor’s format.

Slot
One of the pedal’s memory cells (01-A to 64-D on GP-200).

Working slot, injection slot
The slot the app rewrites for every auditioned preset.

Injection
Writing the preset into the working slot, then selecting it on the pedal.

Bypass
Module turned off (bypassed), the LED is off.

CTRL
A preset “switch” that turns several modules on or off at once.

IR / cab
Simulation of the speaker and the microphone (impulse response).

LUFS / LU
Units of perceived loudness. Default target: −14 LUFS (normal; −16 on GP-150/GP-180), −11 LUFS (lead), adjustable by ±3 LU preset by preset; the tolerance is 0.4 LU.

SysEx
MIDI “system exclusive” messages: they are what carries presets to the pedal. The browser asks for a special permission to use them.

Web MIDI
The browser feature that lets the page talk to the pedal over USB.

Known limits

- Only Gemini, OpenRouter and Anthropic are offered (the other providers refuse calls from a browser).
- The API key is remembered in the browser (box to untick on a shared computer).
- USB sending, volume matching and memory rearranging were verified with a simulated pedal and microphone, byte by byte against the desktop program; reading and listening were also confirmed on a real GP-200 pedal.
- The live chain order (v0.38) relies on a frame decoded from a capture of the Valeton editor (17 moves) and was checked with a simulated pedal that returns the real echo; it still has to be confirmed on your pedal. The patch number appears twice in the frame and was only seen on a single patch. The long press with a finger has only been tested with simulated touch events: it still has to be confirmed on a real phone or tablet.
- Goaltoday's custom firmware: its limits are described at the end of chapter 14.
- The app is a free project, not affiliated with Valeton.